



UNIVERZITET U SARAJEVU

ELEKTROTEHNIČKI FAKULTET SARAJEVO

Odsjek za računarstvo i informatiku

Projektni zadatak

Volunteering management platform

Predmet: Praktikum – Napredne web tehnologije

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Opis

Aplikacija za upravljanje volontiranjem pruža sveobuhvatnu podršku organizatorima i volonterima kroz jednostavno kreiranje i upravljanje volonterskim aktivnostima. Korisnici mogu brzo kreirati naloge, pregledavati dostupne aktivnosti i prijavljivati se putem intuitivnog interfejsa. Organizatori mogu pratiti prisustvo volontera, automatski generisati potvrde o volontiranju i analizirati statistike angažmana. Sistem notifikacija obavještava korisnike o novim događajima i promjenama rasporeda, dok funkcije poput pretrage i filtriranja omogućavaju lak pronalazak aktivnosti. Aplikacija također omogućava kreiranje timova, društvenu integraciju i komunikaciju putem chata ili foruma, čime podstiče zajedništvo i interakciju među volonterima.

Funkcionalnosti

1. **Kreiranje i upravljanje volonterskim aktivnostima** – Organizatori mogu dodavati nove volonterske aktivnosti s opisom, datumom, lokacijom i brojem potrebnih volontera.
2. **Prijava na volonterske aktivnosti** – Volonteri mogu pregledati dostupne aktivnosti i prijaviti se za učešće.
3. **Profil korisnika** – Svaki korisnik ima svoj profil s osnovnim podacima, pregledom prijavljenih i završenih aktivnosti.
4. **Sistem notifikacija** – Obavještavanje korisnika o novim aktivnostima, promjenama u rasporedu i podsjetnicima.
5. **Evidencija prisustva** – Organizatori mogu označiti prisustvo volontera na događajima.
6. **Sistem ocjenjivanja i komentara** – Volonteri mogu ocijeniti aktivnosti i ostaviti komentare kako bi poboljšali buduće događaje.
7. **Pretraga i filtriranje aktivnosti** – Korisnici mogu pretraživati aktivnosti po kategorijama, lokaciji i datumu.
8. **Statistike i izvještaji** – Organizatori mogu vidjeti statistiku volonterskog angažmana (broj prijavljenih, broj završenih aktivnosti itd.).
9. **Sistem uloga i dozvola** – Razlikovanje korisničkih uloga (volonter, organizator, administrator) s odgovarajućim pravima.
10. **Kreiranje timova** – Volonteri se mogu udruživati u timove za zajedničko volontiranje.
11. **Automatsko generisanje potvrda o volontiranju** – Volonteri mogu preuzeti certifikate ili potvrde o učešću u aktivnostima.
12. **Društvena integracija** – Dijeljenje volonterskih aktivnosti na društvenim mrežama.
13. **Sistem preporuka** – Aplikacija može preporučivati aktivnosti na osnovu interesa korisnika.
14. **Chat ili forum za komunikaciju** – Omogućava volonterima i organizatorima da komuniciraju unutar aplikacije.

Korisnički interfejs za funkcionalnosti

1. Kreiranje i upravljanje volonterskim aktivnostima

VolunteerHub

Dashboard

Events

Volunteers

Reports

Help

Create Event

Dashboard > Volunteers > Manage

Volunteer Management

Search volunteers

Search by name, skills or av

Add Volunteer

All Volunteers

Active

Inactive

New Applicants

	Sarah Johnson	Environmental	15 events	4.8/5	Active	  
	Michael Chen	Education	8 events	4.5/5	Active	  
	Jessica Williams	Healthcare	12 events	4.7/5	Active	  
	David Rodriguez	Community	6 events	4.2/5	Active	  

Add New Volunteer

First Name

Last Name

Enter first name

Enter last name

Email

Enter email address

Phone

Enter phone number

Skills & Interests

Enter skills, separated by commas

Availability

Weekdays, weekends, evenings, etc.

Cancel

Add Volunteer

2. Prijava na volonterske aktivnosti

VolunteerHub

Dashboard

Events

Volunteers

Reports

Log In

Sign Up

Home > Volunteer Activities

Available Volunteer Activities

Search activities

Filter

Search by keyword, location

All Activities

Environmental

Education

Community

Healthcare

Environmental

Park Cleanup

Help clean up Central Park this weekend. All supplies provided. 3 hours commitment.

Apply Now

Learn More

Education

Literacy Tutoring

Tutor elementary students in reading and writing. Weekly 2-hour sessions for 3 months.

Apply Now

Learn More

Community

Food Bank Helper

Sort and distribute food at the local food bank. Flexible hours available.

Apply Now

Learn More

Healthcare

Hospital Volunteer

Assist hospital staff with non-medical tasks. Training provided. Background check required.

Apply Now

Learn More

Animal Welfare

Animal Shelter

Help care for animals at the local shelter. Tasks include walking dogs and socializing cats.

Apply Now

Learn More

Senior Care

Senior Companion

Provide companionship to seniors at assisted living facilities. Weekly visits of 1-2 hours.

Apply Now

Learn More

Recently Applied Activities

Beach Cleanup

Environmental

Applied on May 15, 2023

Pending Approval

Homeless Shelter

Community

Applied on May 10, 2023

Approved

Upcoming Volunteer Schedule

Homeless Shelter Assistance

May 25, 2023 • 9:00 AM - 12:00 PM • Downtown Community Center

River Cleanup Project

June 5, 2023 • 8:00 AM - 11:00 AM • Riverside Park

3. Profil korisnika

VolunteerHub

Dashboard

Activities

Community

MessagesFind Activities

Dashboard > User Profile

Profile Information

Manage your personal details and preferences

Edit Profile

Skills & Interests

12 skills added, 5 interest areas

Manage SkillsAdd Interests

Impact Summary

42 volunteer hours, 8 completed activities

View Certificate

Personal Information

Sarah Johnson

Environmental Advocate

sarah.johnson@email.com

Primary Email

+1 (555) 123-4567

Mobile

123 Volunteer St, Portland, OR

Address

Notification Preferences

Email Notifications

Receive activity updates via email

SMS Alerts

Get text reminders for upcoming activities

Activity History

Upcoming Activities

Saved Activities

Completed Activities

Beach Cleanup	Environmental	May 15, 2023	4 hours	☆	
Literacy Program	Education	April 22, 2023	6 hours	☆	
Food Bank Assistance	Community	March 10, 2023	5 hours	☆	
Senior Center Visit	Healthcare	February 5, 2023	3 hours	☆	

Registered Activities

Park Restoration	Environmental	June 12, 2023	Pending		
Homeless Shelter	Community	June 20, 2023	Pending		

View All Activities

4. Sistem notifikacija

VolunteerHub

Activities

Profile

Notifications

Find Activities

Home > Notifications

Notifications

AllUnreadActivitiesRemindersSystem

Search notificationsMark all as read

Today

New volunteer opportunity
Beach Cleanup this Saturday needs 5 more volunteers

Schedule change
Community Garden event moved to 2PM

Activity reminder
Teaching workshop starts tomorrow at 10AM

Yesterday

Impact milestone reached
You've completed 50 volunteer hours!

New skill badge earned
You've earned the 'Team Leader' badge

Earlier this week

Feedback requested
Please rate your experience at Food Bank event

New message
Event coordinator sent you a message

Load more

Notification Preferences
Customize how and when you receive notifications from VolunteerHub

Manage Settings

5. Evidencija prisustva

VolunteerHub

Dashboard

Events

Volunteers

Attendance

Reports

HelpNew Event

Events > Beach Cleanup > Attendance

Beach Cleanup - Attendance Tracking

Saturday, June 15, 2023 • 9:00 AM - 12:00 PM • Ocean Park

Search volunteers

Mark All Present

Enter name or email

All Volunteers (24)Present (18)Absent (6)

Attendance List

✓		Michael Brown	michael.brown@email.com	Present	9:05 AM		
✓		Jessica Lee	jessica.lee@email.com	Present	8:55 AM		
✓		David Wilson	david.wilson@email.com	Present	9:10 AM		
✗		Emily Taylor	emily.taylor@email.com	Absent	-		
✓		Robert Johnson	robert.johnson@email.com	Present	9:02 AM		
✗		Sophia Martinez	sophia.martinez@email.com	Absent	-		

Showing 6 of 24 volunteers

Load More

Attendance Summary



Total Volunteers

24 registered volunteers



Present

18 volunteers (75%)



Absent

6 volunteers (25%)

Export Attendance

Send Reminder

6. Sistem ocjenjivanja i komentara

VolunteerHub

Dashboard

Events

Volunteers

Attendance

Feedback

Reports

Notifications

New Event

Dashboard

Feedback

Activity Ratings

Activity Ratings & Feedback

All FeedbackPending ReviewRespondedArchived

Search feedback

Export Data

Search by event, volunteer

Recent Feedback

☆	Beach Cleanup	Sarah Johnson	4.5/5	2 days ago	Pending Response	👁️ ↶
☆	Food Drive	Michael Chen	5/5	3 days ago	Pending Response	👁️ ↶
☆	Senior Center Visit	Emily Rodriguez	3/5	1 week ago	Responded	👁️ ✔️
☆	Park Restoration	David Kim	4/5	1 week ago	Responded	👁️ ✔️
☆	Homeless Shelter	Jessica Taylor	2/5	2 weeks ago	Escalated	👁️ !

Feedback Details

Beach Cleanup - June 15, 2023

Feedback from Sarah Johnson

The event was well organized and I enjoyed participating. The equipment provided was adequate, but we could have used more trash bags. The team leader was very supportive and encouraging. I would definitely volunteer for this type of event again!

RespondArchive

Overall Rating

4.5/5

Organization

5/5

Equipment

3/5

Leadership

5/5

Would Volunteer Again

Yes

Response

Your response to Sarah

Thank the volunteer and address their feedback...

Send ResponseSave Draft

7. Pretraga i filtriranje aktivnosti

Events > Activities > Search

Find Activities

Search Filters

Narrow down activities by applying filters

Saved Searches

Quick access to your frequent searches

View All

Search Criteria

Keywords

Search by activity name or

Category

Select category

Location

Enter city, state or zip code

Date Range

Select dates

Search Activities

Clear Filters

All Activities

Upcoming

Past

My Favorites

Search Results (24)

Beach Cleanup	Environmental	San Diego	Jun 15, 2023	
Food Drive	Community Service	Los Angeles	Jun 18, 2023	
Senior Center Visit	Elderly Care	San Francisco	Jun 20, 2023	
Park Restoration	Environmental	Portland	Jun 22, 2023	
Homeless Shelter	Community Service	Seattle	Jun 25, 2023	

Load More

8. Statistike i izvještaji

Engagement Statistics

Overview

Volunteers

Activities

Events

Trends

Total

1,248

Registered Volunteers

View Details

Total

356

Completed Activities

View Details

Total

42

Active Events

View Details

Average

12.5 hrs

Volunteer Time per Month

View Details

Volunteer Engagement Trends

Chart visualization would appear here

Recent Activity Completion

Beach Cleanup	Environmental	24 Volunteers	Apr 15, 2023	Completed	
Food Drive	Community Service	18 Volunteers	Apr 12, 2023	Completed	
Senior Center Visit	Elderly Care	12 Volunteers	Apr 10, 2023	Completed	
Park Restoration	Environmental	32 Volunteers	Apr 8, 2023	Completed	
Literacy Program	Education	15 Volunteers	Apr 5, 2023	Completed	

View All Activities

Top Volunteers

Sarah Johnson

42 hours this month

Michael Chen

38 hours this month

Jessica Williams

35 hours this month

David Rodriguez

32 hours this month

Emily Patel

30 hours this month

View All Volunteers

Role Management

Manage user roles and their associated permissions across the platform

Add New Role

Bulk Edit Permissions

System Roles

Custom Roles

Permission Sets

Role Name	Users	Last Modified	Status	Actions
 Administrator	3	Yesterday	Active	 
 Organizer	12	2 days ago	Active	 
 Volunteer	156	1 week ago	Active	 

Role Details



SELECTED ROLE

Administrator

Full access to all system features and settings

Edit Role

View Users



PERMISSION SUMMARY

Full Access

This role has unrestricted access to all 28 system features

Modify Permissions

Permission Matrix

Feature	Administrator	Organizer	Volunteer
 Dashboard	Full Access	View Only	Limited View
 Events	Full Access	Create/Edit	View Only
 Volunteers	Full Access	View/Assign	View Self
 Attendance	Full Access	Record/Edit	View Self
 Feedback	Full Access	View/Respond	Submit Only
 Reports	Full Access	View Only	No Access
 Role Management	Full Access	No Access	No Access

10. Kreiranje timova

Volunteers > Teams > Create Team

Team Creation

Team Details

Create a new volunteer team by filling out the information below.

Team Benefits

Teams allow volunteers to collaborate on activities, build relationships, and track collective impact.

Team Name

Enter team name

Team Description

Describe the team's purpose and goals

Team Category

Select a category

Team Members

Search Volunteers

Search by name or skills

☐	Sarah Johnson	Event Coordination	15 Activities	Available	
☐	Michael Chen	First Aid	8 Activities	Available	
☐	Aisha Patel	Community Outreach	12 Activities	Available	
☐	David Rodriguez	Translation	5 Activities	Available	

Selected Team Members (0)

No team members selected yet. Add volunteers from the list above.

Team Settings

Allow members to invite others
Team members can invite other volunteers to join

Team visibility
Control who can see this team

Select

Team notifications
Send updates to all team members

Create TeamCancel

11. Automatsko generisanje potvrda o volontiranju

Dashboard > Certificates > Generator

Certificate Generator

Create and download certificates for volunteers who participated in activities

Individual Certificate Bulk Generation

Certificate Details

Volunteer Name

Search and select volunteer

Event/Activity

Select event or activity

Start Date

MM/DD/YYYY

End Date

MM/DD/YYYY

Hours Contributed

Enter total hours

Certificate Title

Certificate of Participation

Custom Message

Enter a personalized message for the certificate

Certificate Template



Standard

Classic certificate design with organization logo

Select



Premium

Elegant design with gold accents and seal

Select



Modern

Contemporary design with customizable colors

Select

Signature Options

Signatory Name

Name of person signing certi

Signatory Title

e.g. Program Director

Upload Signature (Optional)

No file chosen

Preview



Certificate Preview

Fill in the details above to see a preview of your certificate

Generate Preview

Download Certificate

Email to Volunteer

Save Template

12. Društvena integracija



Dashboard > Social Media Integration

Social Media Integration

Share volunteer activities and achievements across social media platforms

Platform Settings

Content Templates

Analytics

Connected Platforms



Facebook

Share volunteer events and achievements on your organization's page

Connect

Learn More



Twitter

Post updates and volunteer opportunities to increase engagement

Connect

Learn More



Instagram

Share visual content from volunteer events and activities

Connect

Learn More



LinkedIn

Connect with professionals and share volunteer opportunities

Connect

Learn More



YouTube

Share video content from volunteer events and testimonials

Connect

Learn More



Add Platform

Connect additional social media platforms to expand your reach

Add New

View Options

Sharing Options



Auto-share new events

Automatically post when new volunteer events are created



Share volunteer achievements

Post when volunteers earn certificates or reach milestones



Team accomplishments

Share when volunteer teams complete projects or reach goals



Impact statistics

Share periodic reports on volunteer impact and hours served



Default Sharing Message

Default message template

Check out our volunteer opportunity: {{event_name}} on {{event_date}}! Join us at {{location}} and make a difference. #Volunteering #Community #MakeADifference

Available variables: {{event_name}}, {{event_date}}, {{location}}, {{organization}}, {{volunteer_name}}, {{hours_served}}

Save Settings

Preview Share

13. Sistem preporuka

Recommended Activities

Based on volunteer interests and past participation

For You

Popular

Upcoming

New



High Match

Beach Cleanup

Environmental conservation activity with 85% interest match to volunteer profiles.

View Details

Save



High Match

Animal Shelter

Help at local animal shelter with 78% interest match to volunteer profiles.

View Details

Save



Medium Match

Tutoring Program

Educational support with 65% interest match to volunteer profiles.

View Details

Save

Interest Categories

- Environment
- Education
- Animal Welfare
- Community
- Health

Recently Viewed



Food Bank Distribution

Community service - Last viewed 2 days ago







Park Restoration

Environmental - Last viewed 3 days ago





Volunteer Interests Analysis

 Interest Category	Match Rate	Active Volunteers	Available Activities	
 Environmental	85%	124	8	
 Animal Welfare	78%	96	5	
 Education	65%	87	12	
 Healthcare	52%	63	7	

Recommendation Settings



Interest-Based Recommendations

Show activities based on volunteer interests





Location-Based Recommendations

Show activities near volunteer locations





Skill-Based Recommendations

Show activities matching volunteer skills



14. Chat ili forum za komunikaciju

Community Forum

All Topics

Announcements

General

Event Coordination

Questions

Resources

Search discussions...

🔍

New Discussion

Recent Discussions

- Volunteer orientation for new members

Posted by Sarah Johnson • Announcements • 2h ago • 12 replies
- Beach cleanup coordination for Saturday

Posted by Mike Chen • Event Coordination • 5h ago • 24 replies
- Need help with the mobile check-in process

Posted by Jamie Lee • Questions • 1d ago • 8 replies
- Sharing photos from last week's fundraiser

Posted by Carlos Rodriguez • General • 2d ago • 15 replies
- New training materials for team leaders

Posted by Admin • Resources • 3d ago • 7 replies

Active Channels

Event Planning

Coordinate upcoming events and activities

Join

Team Leaders

Discussion for volunteer team leaders

Join

New Volunteers

Welcome space for new volunteers

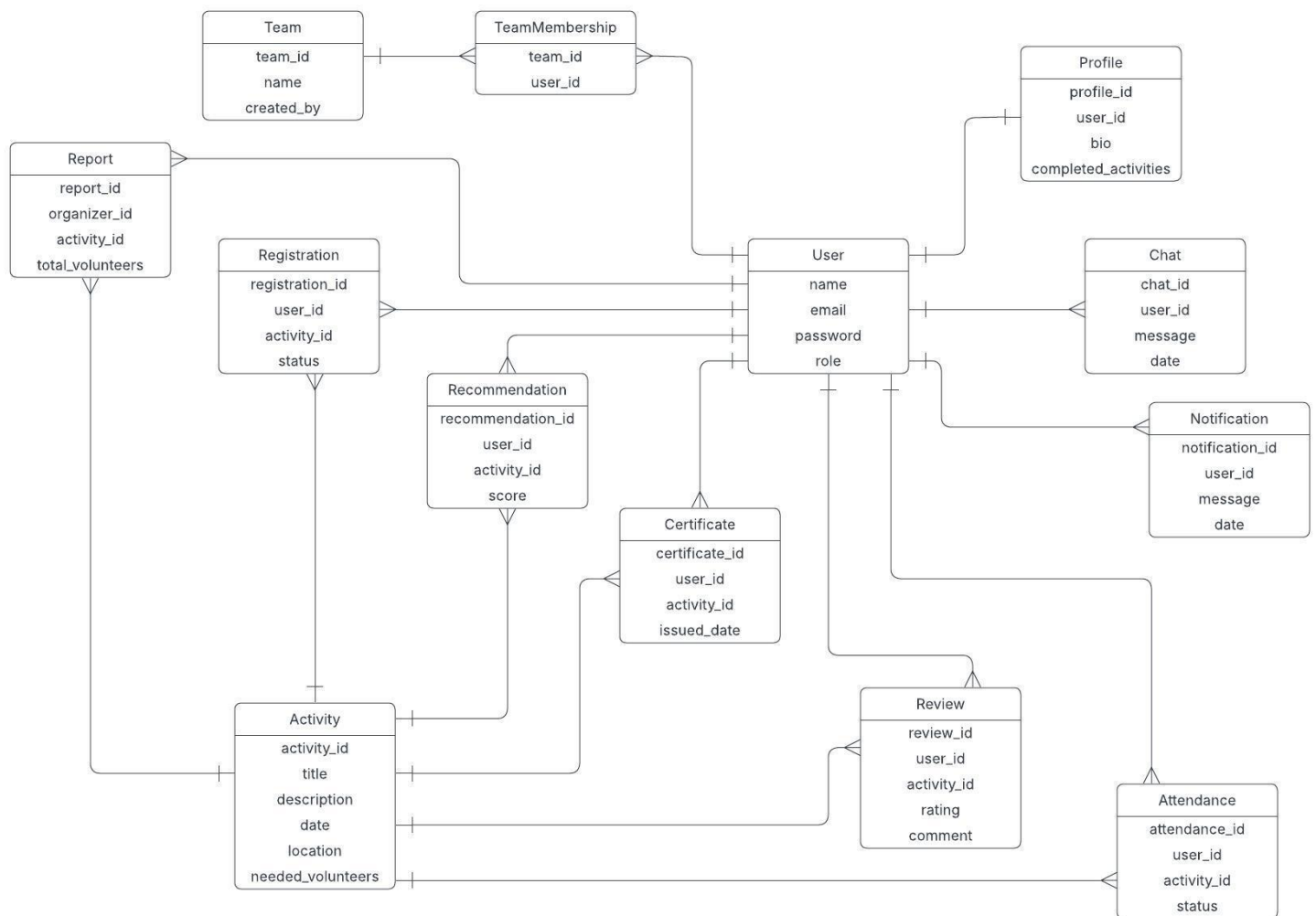
Join

Tech Support

Get help with app and technical issues

Join

ER diagram



Primjeri sličnih aplikacija

- **Volonteka:** Ova platforma omogućava volonterima da pronađu volonterske aktivnosti, prijave se na njih i evidentiraju svoje iskustvo. (<https://volonteka.hr/zagreb-i-okolica>)

- **Volonterium:** Open source aplikacija za upravljanje volontiranjem, razvijena korištenjem Tauri framework-a (Rust backend i Vue.js frontend), koja omogućava administratorima efikasno upravljanje podacima o volonterima, događajima i povratnim informacijama. (<https://github.com/nrastija/Volonterium>)
- **Volonteri.hr:** Inovativni online sistem za brzo i učinkovito pretraživanje ponude volonterskih pozicija i potražnje volontera. (<https://volonteri.hr>)

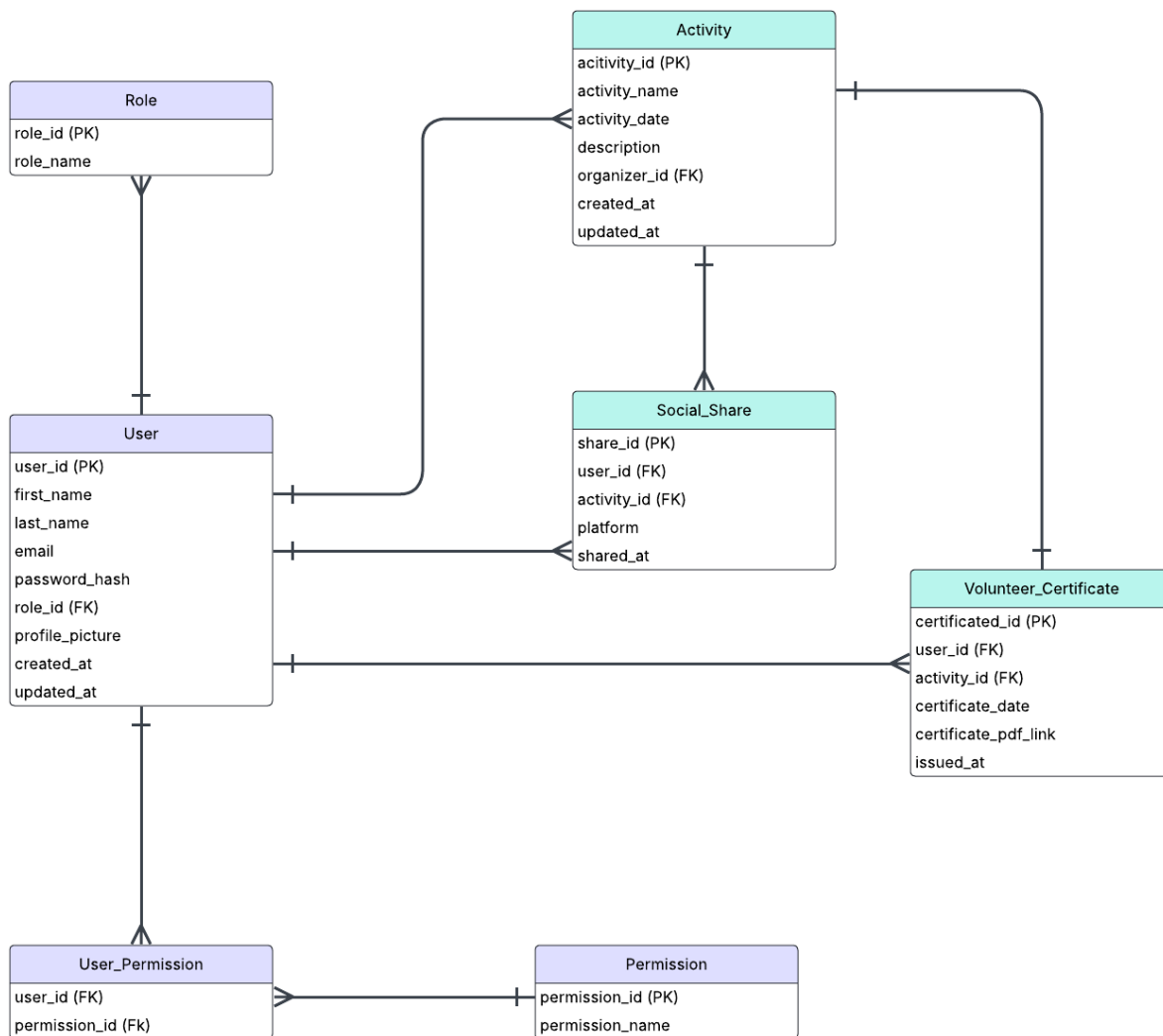
Zadatak 2

Možemo podijeliti aplikaciju na sljedeće mikroservise:

1. User service (Korisnički servis)

- Registracija i prijava korisnika (autentifikacija i autorizacija)
- Upravljanje profilima korisnika
- Sistem uloga i dozvola (volonter, organizator, administrator)
- Automatsko generisanje potvrda o volontiranju
- Društvena integracija (dijeljenje aktivnosti na društvenim mrežama)

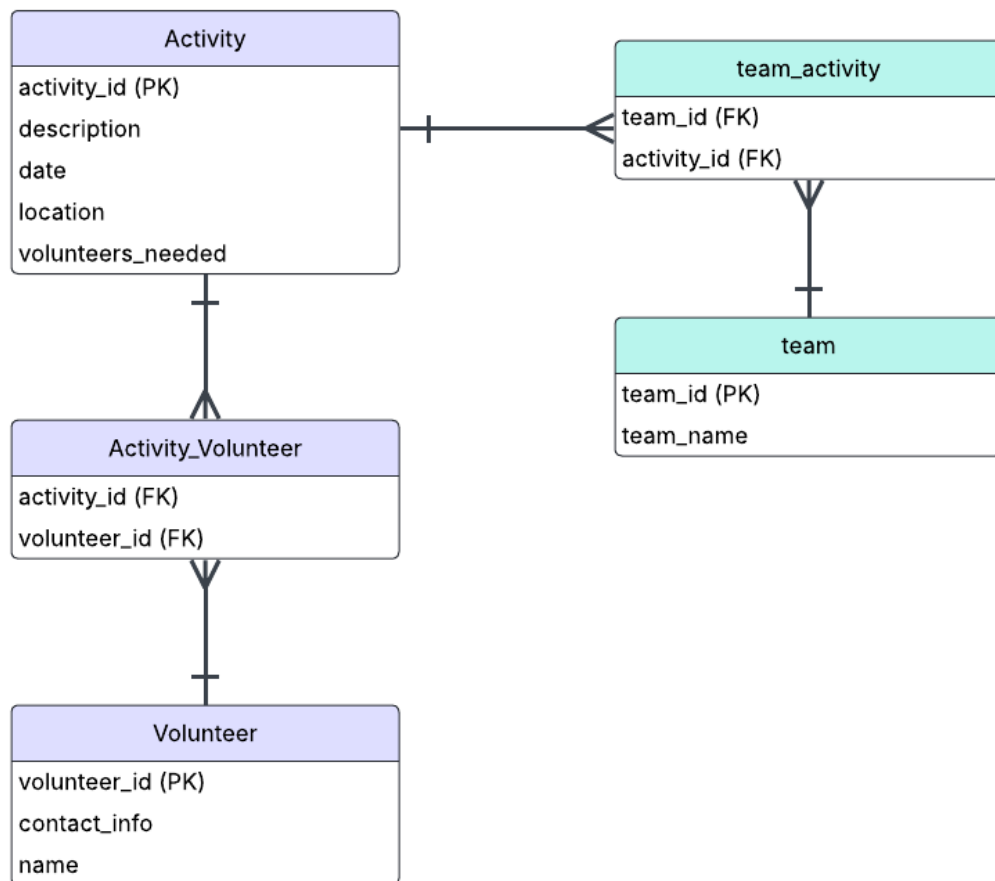
ER dijagram:



2. Activity management service (Upravljanje aktivnostima)

- Kreiranje, uređivanje i brisanje volonterskih aktivnosti
- Upravljanje podacima o aktivnostima (opis, datum, lokacija, broj potrebnih volontera)
- Pretraga i filtriranje aktivnosti
- Kreiranje timova za zajedničko volontiranje

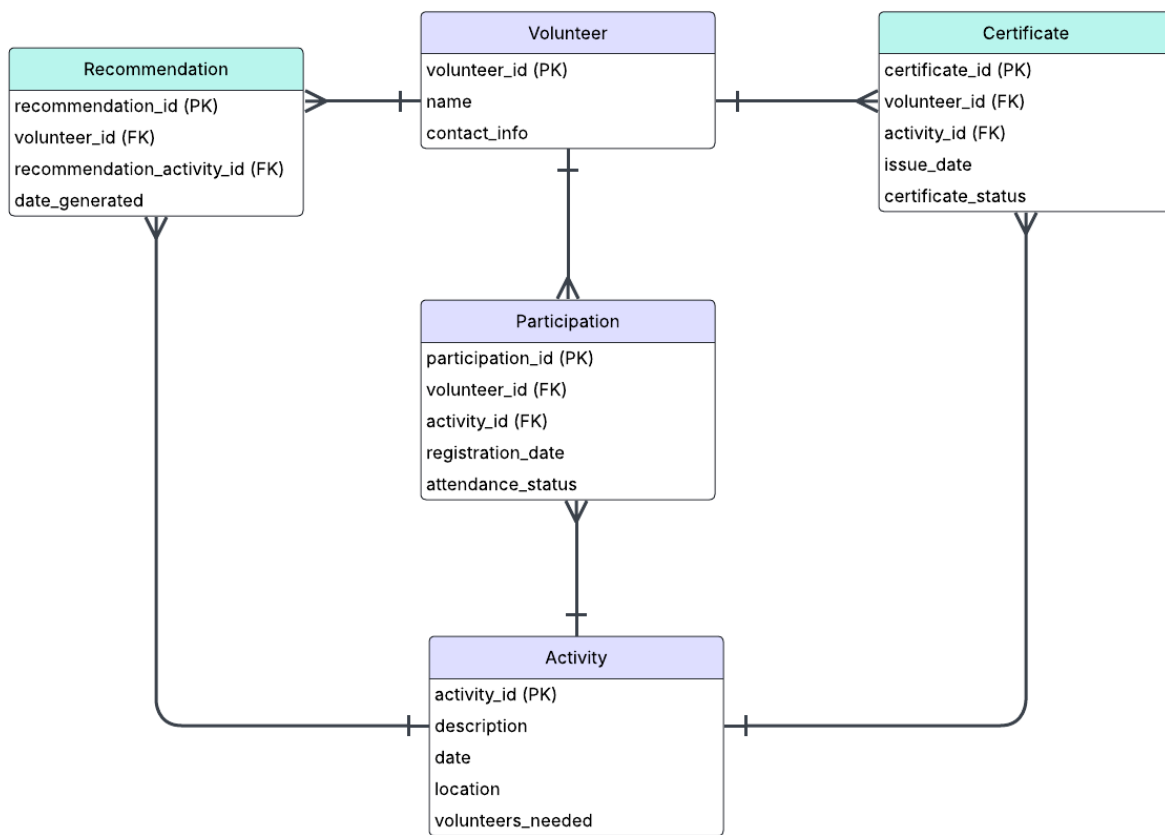
ER dijagram:



3. Participation service (Prijava i prisustvo)

- Prijava volontera na aktivnosti
- Evidencija prisustva volontera
- Automatsko generisanje potvrda o volontiranju
- Sistem preporuka volonterskih aktivnosti na osnovu prethodnih prijava

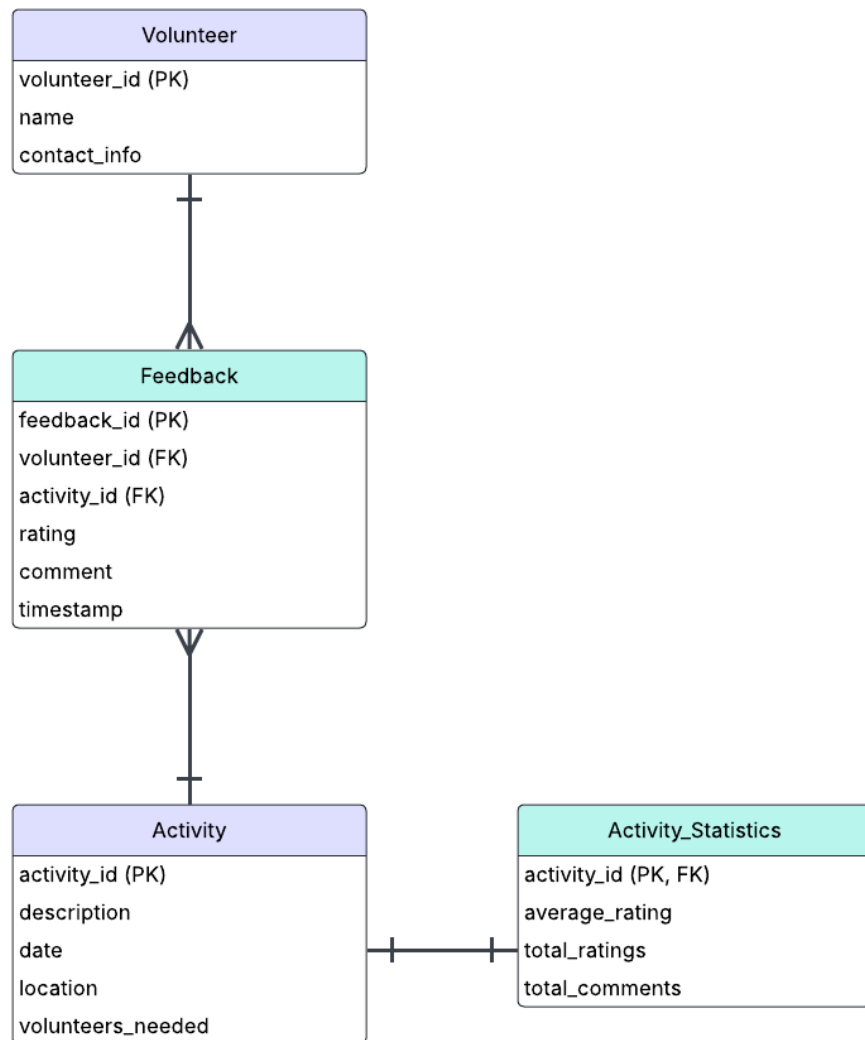
ER dijagram:



4. Feedback & rating service (Ocjene i komentari)

- Sistem ocjenjivanja volonterskih aktivnosti
- Komentarisanje aktivnosti
- Prikaz statistike o aktivnostima na osnovu ocjena i komentara

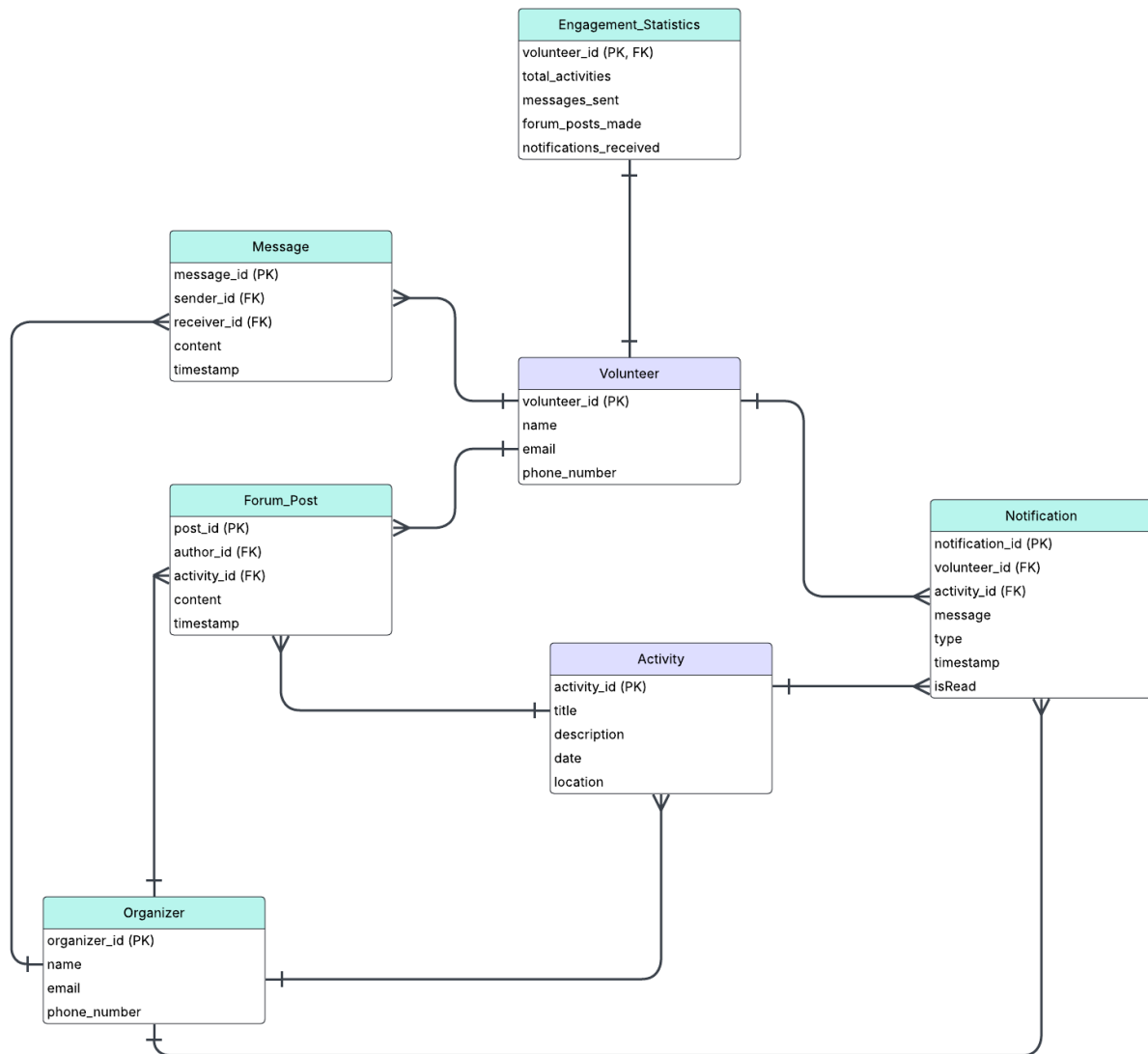
ER dijagram:



5. Notification & communication service (Servis za notifikacije i komunikaciju)

- Sistem notifikacija (nove aktivnosti, promjene rasporeda, podsjetnici)
- Chat ili forum za komunikaciju volontera i organizatora
- Statistike i izvještaji o volonterskom angažmanu

ER dijagram:



Zadatak 3

Activity management service

Dodavanje nove aktivnosti (Create activity):

Uspješan zahtjev:

POST http://localhost:8085/activities

Params Authorization Headers (10) **Body** Scripts Tests Settings Cookies Beautify

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON

```
1 {
2   "activityId": 2,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "Obala rijeke Miljacke",
6   "volunteersNeeded": 15
7 }
```

Body Cookies (1) Headers (5) Test Results 201 Created 3.67 s 303 B

{ } JSON Preview Visualize

```
1 {
2   "activityId": 2,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "Obala rijeke Miljacke",
6   "volunteersNeeded": 15
7 }
```

Neuspješan zahtjev (Description je obavezan):

POST http://localhost:8085/activities

Params Authorization Headers (10) **Body** Scripts Tests Settings Cookies Beautify

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON

```
1 {
2   "activityId": 4,
3   "description": "",
4   "date": "2025-04-25",
5   "location": "Naselje Alipašino Polje",
6   "volunteersNeeded": 8
7 }
```

Body Cookies (1) Headers (4) Test Results 400 Bad Request 2.11 s 268 B

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03 20:29:27",
3   "errorType": "validation_error",
4   "message": "Description is required",
5   "field": "description"
6 }
```

Dobavljanje aktivnosti sa specifičnim ID-em (Get activity by ID):

Uspješan zahtjev:

GET

http://localhost:8085/activities/1

Send

ParamsAuthorizationHeaders (8)BodyScriptsTestsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results

200 OK75 ms298 B

Save Response

{ } JSON

PreviewVisualize

```
1 {
2   "activityId": 1,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "Obala rijeke Miljacke",
6   "volunteersNeeded": 15
7 }
```

Neuspješan zahtjev (Activity not found with that ID):

GET

http://localhost:8085/activities/18

Send

ParamsAuthorizationHeaders (8)BodyScriptsTestsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results

404 Not Found39 ms285 B

Save Response

{ } JSON

PreviewVisualize

```
1 {
2   "timestamp": "2025-04-03 20:38:52",
3   "errorType": "not_found",
4   "message": "Activity with ID 18 not found",
5   "field": null
6 }
```

Dobavljanje svih aktivnosti (Get all activities):

Uspješan zahtjev:

GET http://localhost:8085/activities

Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Body Cookies (1) Headers (5) Test Results 200 OK 31 ms 833 B Save Response

{ } JSON Preview Visualize

```
3      "activityId": 2,
4      "description": "Čišćenje obale rijeke",
5      "date": "2025-04-15",
6      "location": "Obala rijeke Miljacke",
7      "volunteersNeeded": 15
8    },
9    {
10     "activityId": 3,
11     "description": "Dobrovoljno davanje krvi",
12     "date": "2025-04-30",
13     "location": "Crveni križ Sarajevo",
14     "volunteersNeeded": 20
15   },
16   {
17     "activityId": 4,
18     "description": "Dobrovoljno davanje krvi",
19     "date": "2025-04-30",
20     "location": "Crveni križ Sarajevo",
21     "volunteersNeeded": 20
22   },
23   {
24     "activityId": 5,
```

Ažuriranje aktivnosti sa ID-em (Update activity):

Uspješan zahtjev (Number of volunteers changed):

PUT http://localhost:8085/activities/2 Send

Params Authorization Headers (10) Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Beautify

```
1 {
2   "activityId": 2,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "Obala rijeke Miljacke",
6   "volunteersNeeded": 56
7 }
```

Body Cookies (1) Headers (5) Test Results 200 OK • 145 ms • 298 B

{ } JSON Preview Visualize

```
1 {
2   "activityId": 2,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "Obala rijeke Miljacke",
6   "volunteersNeeded": 56
7 }
```

Neuspješan zahtjev (No activity with that ID):

PUT http://localhost:8085/activities/9 Send

Params Authorization Headers (10) Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Beautify

```
1 {
2   "activityId": 2,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "Obala rijeke Miljacke",
6   "volunteersNeeded": 56
7 }
```

Body Cookies (1) Headers (5) Test Results 404 Not Found • 130 ms • 284 B

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03 20:48:23",
3   "errorType": "not_found",
4   "message": "Activity with ID 9 not found",
5   "field": null
6 }
```

(Location is required):

PUT http://localhost:8085/activities/2 Send

Params Authorization Headers (10) Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Beautify

```
1 {
2   "activityId": 2,
3   "description": "Čišćenje obale rijeke",
4   "date": "2025-04-15",
5   "location": "",
6   "volunteersNeeded": 56
7 }
```

Body Cookies (1) Headers (4) Test Results 400 Bad Request 39 ms 262 B

{ JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03 20:49:43",
3   "errorType": "validation_error",
4   "message": "Location is required",
5   "field": "location"
6 }
```

Brisanje aktivnosti (Delete activity with ID):

Uspješan zahtjev (Activity deleted):

DELETE http://localhost:8085/activities/1 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	Bulk Edit
	Key	Value	Description	

Body Cookies (1) Headers (3) Test Results 204 No Content 256 ms 112 B

Raw Preview Visualize

1

GET

http://localhost:8085/activities/1

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	*** Bulk Edit
	Key	Value	Description	

BodyCookies (1)Headers (5)Test Results404 Not Found16 ms284 BSave Response

{ } JSON

PreviewVisualize

1 {

2 "timestamp": "2025-04-03 20:54:09",

3 "errorType": "not_found",

4 "message": "Activity with ID 1 not found",

5 "field": null

6 }

Neuspješan zahtjev (No activity with that ID):

DELETE

http://localhost:8085/activities/15

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	*** Bulk Edit
	Key	Value	Description	

BodyCookies (1)Headers (5)Test Results404 Not Found2.02 s285 B

{ } JSON

PreviewVisualize

1 {

2 "timestamp": "2025-04-03 20:51:50",

3 "errorType": "not_found",

4 "message": "Activity with ID 15 not found",

5 "field": null

6 }

Dohvatanje svih volontera (Get all volunteers):

Uspješan zahtjev:

The screenshot displays a REST client interface with the following components:

- Request Bar:** Method **GET**, URL **http://localhost:8085/volunteers**, and a **Send** button.
- Navigation Tabs:** Params, Authorization, Headers (8), Body, Scripts, Settings, and Cookies.
- Query Params Table:**

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		
- Response Section:**
 - Body Tab:** Selected, showing a **200 OK** status, **173 ms** duration, and **295 B** size.
 - JSON View:** Shows the response body as a JSON array of two volunteer objects.

```
1  [  
2    {  
3      "volunteerId": 1,  
4      "name": "Ana",  
5      "contactInfo": "ana@example.com"  
6    },  
7    {  
8      "volunteerId": 2,  
9      "name": "Marko",  
10     "contactInfo": "marko@example.com"  
11   }  
12 ]
```

Dohvatanje volontera sa ID-em (Get volunteer by ID):

Uspješan zahtjev:

GET

▼

http://localhost:8085/volunteers/2

Send

▼

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (5)

Test Results

200 OK

110 ms

230 B

Save Response

...

{} JSON

▼

▶ Preview

Visualize

▼

```
1 {
2   "volunteerId": 2,
3   "name": "Marko",
4   "contactInfo": "marko@example.com"
5 }
```

Neuspješan zahtjev:

GET

▼

http://localhost:8085/volunteers/3

Send

▼

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (5)

Test Results

404 Not Found

131 ms

285 B

Save Response

...

{} JSON

▼

▶ Preview

Visualize

▼

```
1 {
2   "timestamp": "2025-04-03 21:21:32",
3   "errorType": "not_found",
4   "message": "Volunteer with ID 3 not found",
5   "field": null
6 }
```

Dodavanje novog volontera (Create new volunteer):

Uspješan zahtjev:

POST http://localhost:8085/volunteers

Send

Params Authorization Headers (10) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ☐ Beautify

```
1 {
2   "name": "Sanja Petrović",
3   "contactInfo": "sanjap@example.com"
4 }
5
```

Body Cookies (1) Headers (5) Test Results | ⌚

200 OK • 561 ms • 241 B • 🌐 ⋮

{ } JSON ☐ Preview 🔄 Visualize ☐

```
1 {
2   "volunteerId": 3,
3   "name": "Sanja Petrović",
4   "contactInfo": "sanjap@example.com"
5 }
```

Neuspješan zahtjev:

POST http://localhost:8085/volunteers

Send

Params Authorization Headers (10) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ☐ Beautify

```
1 {
2   "name": "Sanja Petrović",
3   "contactInfo": ""
4 }
5
```

Body Cookies (1) Headers (4) Test Results | ⌚

400 Bad Request • 122 ms • 269 B • 🌐 ⋮

{ } JSON ☐ Preview 🔄 Visualize ☐

```
1 {
2   "timestamp": "2025-04-03 21:26:16",
3   "errorType": "validation_error",
4   "message": "Contact info is required",
5   "field": "contactInfo"
6 }
```

Brisanje volontera (delete volunteer):

Uspješan zahtjev:

DELETE

http://localhost:8085/volunteers/4

Send

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (3)

Test Results

204 No Content

39 ms

112 B

Raw

Preview

Visualize

1

GET

http://localhost:8085/volunteers/4

Send

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (5)

Test Results

404 Not Found

17 ms

285 B

Save Response

JSON

Preview

Visualize

1

2

3

4

5

6

{

"timestamp": "2025-04-03 21:28:44",

"errorType": "not_found",

"message": "Volunteer with ID 4 not found",

"field": null

}

Neuspješan zahtjev:

DELETE

http://localhost:8085/volunteers/6

Send

Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (5)

Test Results

404 Not Found

35 ms

285 B

Save Response

JSON

Preview

Visualize

```

1  {
2    "timestamp": "2025-04-03 21:30:12",
3    "errorType": "not_found",
4    "message": "Volunteer with ID 6 not found",
5    "field": null
6  }

```

Dohvatanje svih timova (Get all teams):

GET

http://localhost:8085/teams

Send

Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
--	-----	-------	-------------	-----	-----------

Body

Cookies (1)

Headers (5)

Test Results

200 OK

55 ms

704 B

Save Response

JSON

Preview

Visualize

```

1  [
2    {
3      "teamId": 1,
4      "teamName": "Zeleni Tim",
5      "teamActivities": [
6        {
7          "id": 2,
8          "activity": {
9            "activityId": 3,
10           "description": "Dobrovoljno davanje krvi",
11           "date": "2025-04-30",
12           "location": "Crveni križ Sarajevo",
13           "volunteersNeeded": 20,
14           "activityVolunteers": []
15         }
16       ]
17     },
18   ]

```

Postbot
Runner
Start Proxy
Cookies
Vault
Trash

Dohvatanje tima sa ID-em (Get team by ID):

Uspješan zahtjev:

GET

http://localhost:8085/teams/1

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results🕒

200 OK • 27 ms • 398 B • 🌐 📄 Save Response ...

{ } JSON ▾▶ Preview🔗 Visualize ▾

🔗 📄 🔍 🔗

```
1 {
2   "teamId": 1,
3   "teamName": "Zeleni Tim",
4   "teamActivities": [
5     {
6       "id": 2,
7       "activity": {
8         "activityId": 3,
9         "description": "Dobrovoljno davanje krvi",
10        "date": "2025-04-30",
11        "location": "Crveni križ Sarajevo",
12        "volunteersNeeded": 20,
13        "activityVolunteers": []
14      }
15    }
16  ]
17 }
```

Neuspješan zahtjev:

GET

http://localhost:8085/teams/7

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results🕒

404 Not Found • 30 ms • 280 B • 🌐 📄 Save Response ...

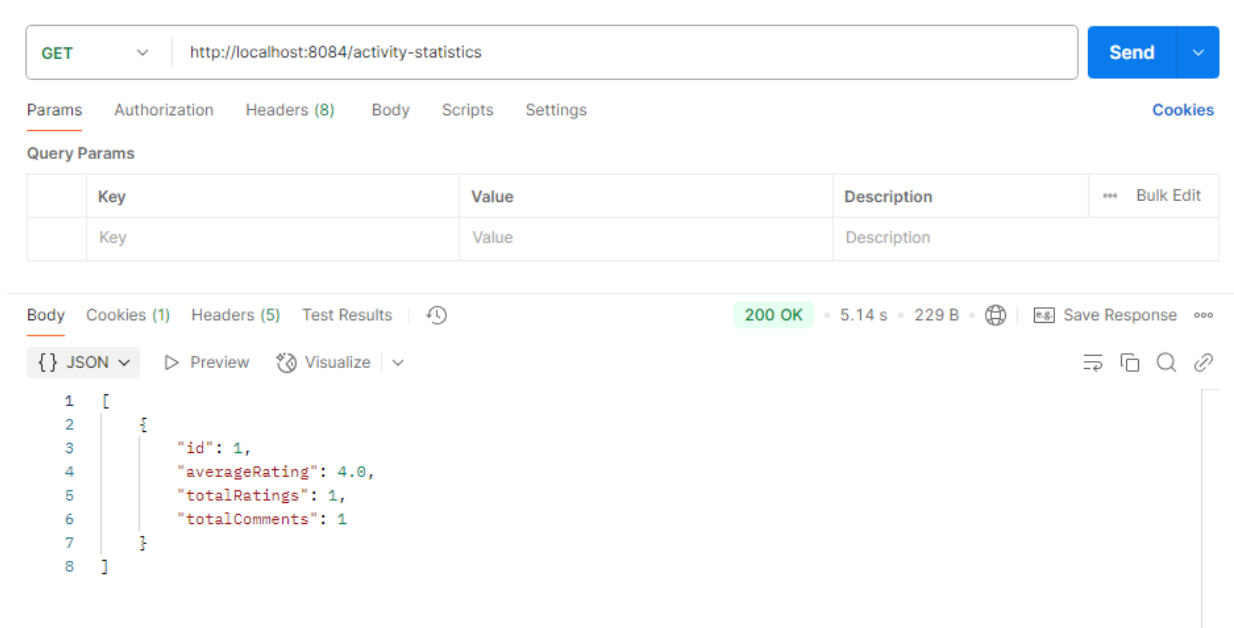
{ } JSON ▾▶ Preview🔗 Visualize ▾

🔗 📄 🔍 🔗

```
1 {
2   "timestamp": "2025-04-03 21:33:48",
3   "errorType": "not_found",
4   "message": "Team with ID 7 not found",
5   "field": null
6 }
```

Feedback service

Dohvatanje svih statistika aktivnosti (Get all activity statistics):



GET http://localhost:8084/activity-statistics

Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

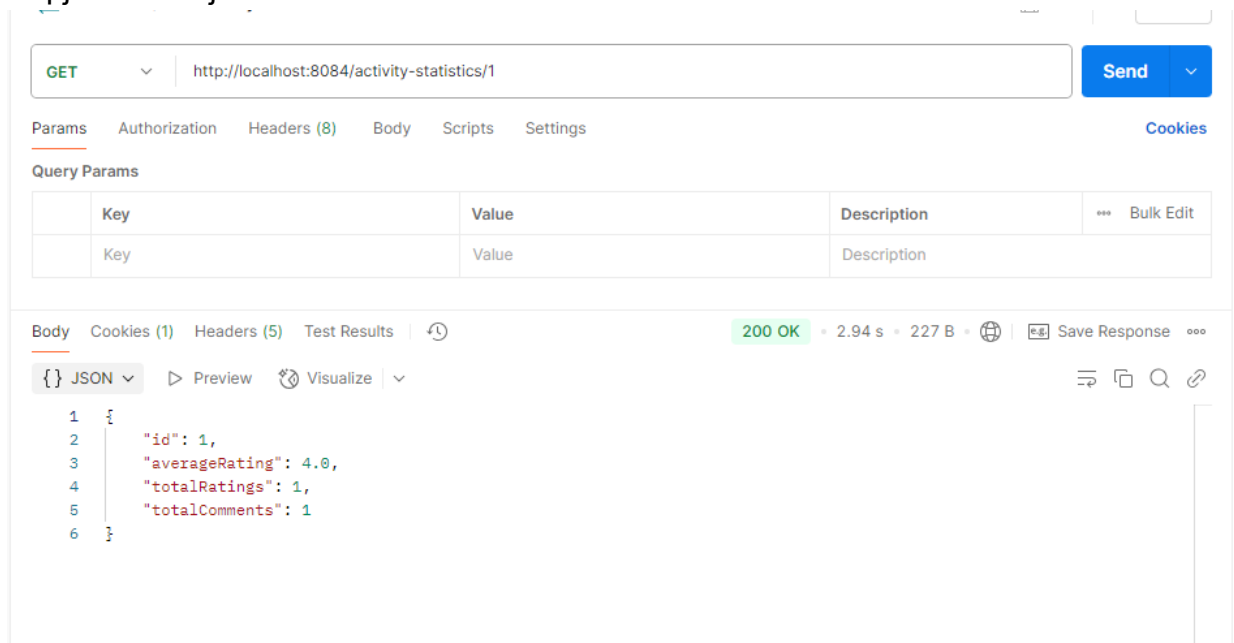
Body Cookies (1) Headers (5) Test Results 200 OK • 5.14 s • 229 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "id": 1,
4     "averageRating": 4.0,
5     "totalRatings": 1,
6     "totalComments": 1
7   }
8 ]
```

Dohvatanje statistike aktivnosti sa ID-em (Get activity statistics by ID):

Uspješan zahtjev:



GET http://localhost:8084/activity-statistics/1

Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 2.94 s • 227 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 1,
3   "averageRating": 4.0,
4   "totalRatings": 1,
5   "totalComments": 1
6 }
```

Neuspješan zahtjev:

GET http://localhost:8084/activity-statistics/2 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 404 Not Found 229 ms 312 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T21:58:31.8245009",
3   "errorType": "Resource Not Found",
4   "message": "Activity Statistics not found with id 2",
5   "field": "id"
6 }
```

Kreiranje novog feedback-a (Create feedback):

Uspješan zahtjev:

POST http://localhost:8084/feedbacks Send

Params Authorization Headers (10) Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Beautify

```
1 {
2   "rating": 5,
3   "comment": "Great work!",
4   "volunteer": {
5     "volunteerId": 1
6   },
7   "activity": {
8     "activityId": 1
9   }
10 }
11
```

Body Cookies (1) Headers (5) Test Results 201 Created 1.42 s 237 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "feedbackId": 2,
3   "rating": 5,
4   "comment": "Great work!",
5   "timestamp": null
6 }
```

Dohvatanje svih aktivnosti (Get all feedbacks):

GET http://localhost:8084/feedbacks Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 150 ms • 552 B • Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "feedbackId": 1,
4     "rating": 4,
5     "comment": "Pomoć bila izuzetno korisna.",
6     "timestamp": "2025-04-03T19:48:58.806+00:00"
7   },
8   {
9     "feedbackId": 2,
10    "rating": 5,
11    "comment": "Great work!",
12    "timestamp": null
13  },
14  {
15    "feedbackId": 3,
```

Dohvatanje feedback-a sa ID-em (get feedback by ID):

Uspješan zahtjev:

GET http://localhost:8084/feedbacks/4 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 940 ms • 231 B • Save Response

{ } JSON Preview Visualize

```
1 {
2   "feedbackId": 4,
3   "rating": 4,
4   "comment": "Great work",
5   "timestamp": null
6 }
```

Neuspješan zahtjev:

GET http://localhost:8084/feedbacks/10 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 404 Not Found • 146 ms • 200 B • Save Response

Raw Preview Visualize

1 Feedback not found with id 10

Notification-communication service

Dohvatanje svih engagement statistika (Get all engagement statistics):

GET http://localhost:8083/engagement-statistics Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 1.04 s • 353 B • Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "id": 1,
4     "totalActivities": 10,
5     "messagesSent": 50,
6     "forumPostsMade": 5,
7     "notificationsReceived": 20
8   },
9   {
10    "id": 2,
11    "totalActivities": 15,
12    "messagesSent": 80,
13    "forumPostsMade": 8,
14    "notificationsReceived": 30
15  }
```

Dohvatanje engagement statistics sa ID-em (Get engagement statistics with ID):

Uspješan zahtjev:

GET http://localhost:8083/engagement-statistics/2

Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 200 OK • 729 ms • 257 B Save Response

JSON Preview Visualize

```
1 {
2   "id": 2,
3   "totalActivities": 15,
4   "messagesSent": 80,
5   "forumPostsMade": 8,
6   "notificationsReceived": 30
7 }
```

Neuspješan zahtjev:

GET http://localhost:8083/engagement-statistics/7

Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 404 Not Found • 1.99 s • 314 B Save Response

JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T22:14:21.9521026",
3   "errorType": "Resource Not Found",
4   "message": "Engagement Statistics not found with id 7",
5   "field": "id"
6 }
```

Dohvatanje svih forum posts (Get all forum posts):

GET http://localhost:8083/forum-posts Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 723 ms • 371 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "postId": 1,
4     "content": "How many people are going to be there?",
5     "timestamp": "2025-04-03T20:11:40.205+00:00"
6   },
7   {
8     "postId": 2,
9     "content": "When does the cleanup start?",
10    "timestamp": "2025-04-03T20:11:40.214+00:00"
11  }
12 ]
```

Dohvatanje forum posts sa ID-em (Get forum post by ID):

Uspješan zahtjev:

GET http://localhost:8083/forum-posts/2 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 179 ms • 261 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "postId": 2,
3   "content": "When does the cleanup start?",
4   "timestamp": "2025-04-03T20:11:40.214+00:00"
5 }
```

Neuspješan zahtjev:

GET http://localhost:8083/forum-posts/6 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 404 Not Found 214 ms 302 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T22:18:40.1956705",
3   "errorType": "Resource Not Found",
4   "message": "ForumPost not found with id 6",
5   "field": "id"
6 }
```

Dohvatanje svih messages (Get all messages):

GET http://localhost:8083/messages Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 200 OK 282 ms 395 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "messageId": 1,
4     "content": "Hey Jane, are you joining the Community Cleanup tomorrow?",
5     "timestamp": "2025-04-03T20:11:40.218+00:00"
6   },
7   {
8     "messageId": 2,
9     "content": "Yes, I'll be there at 9 AM!",
10    "timestamp": "2025-04-03T20:11:40.226+00:00"
11  }
12 ]
```

Dohvatanje message sa ID-em (Get message by ID):

Uspješan zahtjev:

GET http://localhost:8083/messages/2 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 280 ms • 263 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "messageId": 2,
3   "content": "Yes, I'll be there at 9 AM!",
4   "timestamp": "2025-04-03T20:11:40.226+00:00"
5 }
```

Neuspješan zahtjev:

GET http://localhost:8083/messages/5 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 404 Not Found • 280 ms • 300 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T22:29:34.6356285",
3   "errorType": "Resource Not Found",
4   "message": "Message not found with id 5",
5   "field": "id"
6 }
```

Dohvatanje svih notifikacija (Get all notifications):

GET http://localhost:8083/notifications Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 207 ms • 485 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "notificationId": 1,
4     "message": "Activity reminder: Community Cleanup tomorrow at 9 AM.",
5     "type": "Reminder",
6     "timestamp": "2025-04-03T20:11:40.192+00:00",
7     "read": false
8   },
9   {
10    "notificationId": 2,
11    "message": "New forum post: Fundraising event announced!",
12    "type": "Announcement",
13    "timestamp": "2025-04-03T20:11:40.201+00:00",
14    "read": false
15  }
16 ]
```

Dohvatanje notifikacije sa ID-em (Get notification by ID):

Uspješan zahtjev:

GET http://localhost:8083/notifications/1 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 411 ms • 326 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "notificationId": 1,
3   "message": "Activity reminder: Community Cleanup tomorrow at 9 AM.",
4   "type": "Reminder",
5   "timestamp": "2025-04-03T20:11:40.192+00:00",
6   "read": false
7 }
```

Neuspješan zahtjev:

GET http://localhost:8083/notifications/8 Send

Params Authorization Headers (8) Body Scripts Settings Co

Query Params

Key	Value	Description	...	Bulk I
Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 404 Not Found 160 ms 203 B Save Response

Raw Preview Visualize

1 Notification with ID 8 not found

Dohvaćanje svih organizatora (Get all organizers):

GET http://localhost:8083/organizers Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK 199 ms 387 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "organizerId": 1,
4     "name": "John's Volunteer Group",
5     "email": "johnsgroup@example.com",
6     "phoneNumber": "1234567890"
7   },
8   {
```

Postbot Runner Start Proxy Cookies Vault Trash

Dohvaćanje organizatora sa ID-em (Get organizer by ID):

Uspješan zahtjev:

GET

http://localhost:8083/organizers/1

Send

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (5)

Test Results

200 OK

224 ms

273 B

Save Response

{}

JSON

Preview

Visualize

```
1 {
2   "organizerId": 1,
3   "name": "John's Volunteer Group",
4   "email": "johnsgroup@example.com",
5   "phoneNumber": "1234567890"
6 }
```

Neuspješan zahtjev:

GET http://localhost:8083/organizers/9 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	...	Bulk Edit
Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 404 Not Found 637 ms 200 B Save Response

Raw Preview Visualize

1 Organizer not found with ID 9

Participation service

Dohvaćanje svih učešća (Get all participations):

GET http://localhost:8081/participations Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Body Cookies (1) Headers (5) Test Results 200 OK 863 ms 373 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "participationId": 1,
4     "registrationDate": "2025-04-03T20:41:17.794+00:00",
5     "attendanceStatus": "Confirmed"
6   },
7   {
8     "participationId": 2,
9     "registrationDate": "2025-04-03T20:41:17.804+00:00",
10    "attendanceStatus": "Confirmed"
11  }
12 ]
```

Dohvaćanje učešća sa ID-em (Get participation by ID):

Uspješan zahtjev:

GET

http://localhost:8081/participations/1

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results

200 OK2.17 s267 B

{ } JSON

PreviewVisualize

```
1 {
2   "participationId": 1,
3   "registrationDate": "2025-04-03T20:41:17.794+00:00",
4   "attendanceStatus": "Confirmed"
5 }
```

Neuspješan zahtjev:

GET

http://localhost:8081/participations/8

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results

404 Not Found232 ms306 B

{ } JSON

PreviewVisualize

```
1 {
2   "timestamp": "2025-04-03T22:43:30.1886275",
3   "errorType": "Resource Not Found",
4   "message": "Participation not found with id 8",
5   "field": "id"
6 }
```

PostbotRunnerStart ProxyCookiesVaultTrash

Dohvaćanje svih certifikata (Get all certificates):

GET http://localhost:8081/certificates Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
-----	-------	-------------	-----------

Body Cookies (1) Headers (5) Test Results 200 OK 95 ms 351 B Save Response

JSON Preview Visualize

```
1 [
2   {
3     "certificateId": 1,
4     "issueDate": "2025-04-03T20:41:17.821+00:00",
5     "certificateStatus": "Issued"
6   },
7   {
8     "certificateId": 2,
9     "issueDate": "2025-04-03T20:41:17.830+00:00",
10    "certificateStatus": "Issued"
11  }
12 ]
```

Dohvaćanje certifikata sa ID-em (Get certificate by ID):

Uspješan zahtjev:

GET http://localhost:8081/certificates/1 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 200 OK 134 ms 256 B Save Response

JSON Preview Visualize

```
1 {
2   "certificateId": 1,
3   "issueDate": "2025-04-03T20:41:17.821+00:00",
4   "certificateStatus": "Issued"
5 }
```

Neuspješan zahtjev:

GET http://localhost:8081/certificates/9 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 404 Not Found 551 ms 303 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T22:46:36.249918",
3   "errorType": "Resource Not Found",
4   "message": "Certificate not found with id 9",
5   "field": "id"
6 }
```

User service

Dohvaćanje svih rola (Get all roles):

GET http://localhost:8082/roles Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 200 OK 892 ms 197 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "roleId": 1,
4     "roleName": "Admin"
5   }
6 ]
```

Dohvaćanje rola sa ID-em (Get role by ID):

Uspješan zahtjev:

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** http://localhost:8082/roles/1
- Status:** 200 OK
- Response Time:** 752 ms
- Response Size:** 195 B
- Body:** JSON format showing a successful response for role ID 1 with role name "Admin".

Key	Value	Description
Key	Value	Description

```
1 {
2   "roleId": 1,
3   "roleName": "Admin"
4 }
```

Neuspješan zahtjev:

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** http://localhost:8082/roles/9
- Status:** 404 Not Found
- Response Time:** 106 ms
- Response Size:** 297 B
- Body:** JSON format showing an error response: "Resource Not Found".

Key	Value	Description
Key	Value	Description

```
1 {
2   "timestamp": "2025-04-03T22:51:49.5255352",
3   "errorType": "Resource Not Found",
4   "message": "Role not found with id 9",
5   "field": "id"
6 }
```

Dodavanje nove role (Create role):

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:8082/roles
- Body:** A JSON object:

```
{  "roleName": "Volunteer"}
```
- Response:** 201 Created, 4.59 s, 204 B. The response body is a JSON object:

```
{  "roleId": 2,  "roleName": "Volunteer"}
```

Brisanje role (Delete role):

The screenshot shows a REST client interface with the following details:

- Method:** DELETE
- URL:** http://localhost:8082/roles/2
- Body:** A JSON object:

```
{  "roleName": "Volunteer"}
```
- Response:** 204 No Content, 4.00 s, 112 B.

GET http://localhost:8082/roles/2 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies (1) Headers (5) Test Results 404 Not Found 201 ms 297 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T22:59:57.1996067",
3   "errorType": "Resource Not Found",
4   "message": "Role not found with id 2",
5   "field": "id"
6 }
```

Promjena role (Update role):

PUT http://localhost:8082/roles/1 Send

Params Authorization Headers (10) Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Beautify

```
1 {
2   "roleName": "Volunteer"
3 }
```

Body Cookies (1) Headers (5) Test Results 200 OK 821 ms 199 B

{ } JSON Preview Visualize

```
1 {
2   "roleId": 1,
3   "roleName": "Volunteer"
4 }
```

Dohvaćanje svih permisija (Get all permissions):

GET http://localhost:8082/permissions Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK 1.06 s 674 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "permissionId": 1,
4     "permissionName": "READ_USERS",
5     "userPermissions": [
6       {
7         "id": 1,
8         "createdAt": "2025-04-03T20:49:45.855+00:00",
9         "updatedAt": "2025-04-03T20:49:45.855+00:00"
10      }
11    ],
12    "createdAt": "2025-04-03T20:49:45.846+00:00",
13    "updatedAt": "2025-04-03T20:49:45.846+00:00"
14  },
15 ]
```

Dohvaćanje permisije sa ID-em (Get permission by ID):

Uspješan zahtjev:

GET http://localhost:8082/permissions/1 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 233 ms • 417 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "permissionId": 1,
3   "permissionName": "READ_USERS",
4   "userPermissions": [
5     {
6       "id": 1,
7       "createdAt": "2025-04-03T20:49:45.855+00:00",
8       "updatedAt": "2025-04-03T20:49:45.855+00:00"
9     }
10  ],
11  "createdAt": "2025-04-03T20:49:45.846+00:00",
12  "updatedAt": "2025-04-03T20:49:45.846+00:00"
13 }
```

Neuspješan zahtjev:

GET http://localhost:8082/permissions/9 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 404 Not Found • 1.22 s • 303 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T23:04:44.1094408",
3   "errorType": "Resource Not Found",
4   "message": "Permission not found with id 9",
5   "field": "id"
6 }
```

Dohvaćanje svih usera (Get all users):

GET

http://localhost:8082/users

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

BodyCookies (1)Headers (5)Test Results

200 OK • 1.10 s • 384 B • Save Response

{ } JSON

PreviewVisualize

```
1  [
2    {
3      "userId": 1,
4      "firstName": "John",
5      "lastName": "Doe",
6      "email": "john.doe@example.com",
7      "profilePicture": "profilePicUrl",
8      "roleName": "Admin",
9      "permissions": [
10       "READ_USERS",
11       "EDIT_EVENTS"
12     ],
13     "createdAt": "2025-04-03T20:49:45.804+00:00"
14   }
15 ]
```

Dohvaćanje usera sa ID-em (Get user by ID):

Uspješan zahtjev:

GET http://localhost:8082/users/1 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK • 426 ms • 382 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "userId": 1,
3   "firstName": "John",
4   "lastName": "Doe",
5   "email": "john.doe@example.com",
6   "profilePicture": "profilePicUrl",
7   "roleName": "Admin",
8   "permissions": [
9     "READ_USERS",
10    "EDIT_EVENTS"
11  ],
12  "createdAt": "2025-04-03T20:49:45.804+00:00"
13 }
```

Neuspješan zahtjev:

GET http://localhost:8082/users/2 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 404 Not Found • 932 ms • 297 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "timestamp": "2025-04-03T23:07:55.6323298",
3   "errorType": "Resource Not Found",
4   "message": "User not found with id 2",
5   "field": "id"
6 }
```

Dohvaćanje svih podjela na mrežama (Get all social share):

GET http://localhost:8082/social-shares Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK 1.46 s 332 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "shareId": 1,
4     "platform": "Facebook",
5     "sharedAt": "2025-04-03T20:49:46.011+00:00",
6     "createdAt": "2025-04-03T20:49:46.012+00:00",
7     "updatedAt": "2025-04-03T20:49:46.012+00:00"
8   }
9 ]
```

Dohvaćanje social share sa ID-em (Get social share by ID):

Uspješan zahtjev:

GET http://localhost:8082/social-shares/1 Send

Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies (1) Headers (5) Test Results 200 OK 286 ms 330 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "shareId": 1,
3   "platform": "Facebook",
4   "sharedAt": "2025-04-03T20:49:46.011+00:00",
5   "createdAt": "2025-04-03T20:49:46.012+00:00",
6   "updatedAt": "2025-04-03T20:49:46.012+00:00"
7 }
```

Neuspješan zahtjev:

GET

http://localhost:8082/social-shares/9

Send

ParamsAuthorizationHeaders (8)BodyScriptsSettingsCookies

Query Params

	Key	Value	Description	*** Bulk Edit
	Key	Value	Description	

BodyCookies (1)Headers (5)Test Results404 Not Found479 ms305 BSave Response

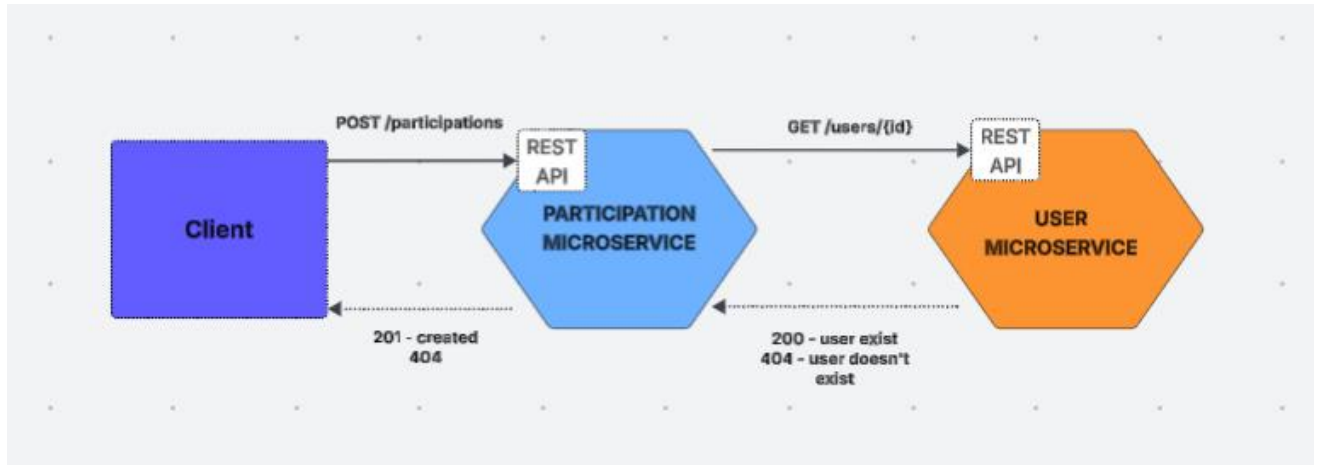
{ } JSONPreviewVisualize

```
1 {
2   "timestamp": "2025-04-03T23:20:11.6681488",
3   "errorType": "Resource Not Found",
4   "message": "Social share not found with id 9",
5   "field": "id"
6 }
```

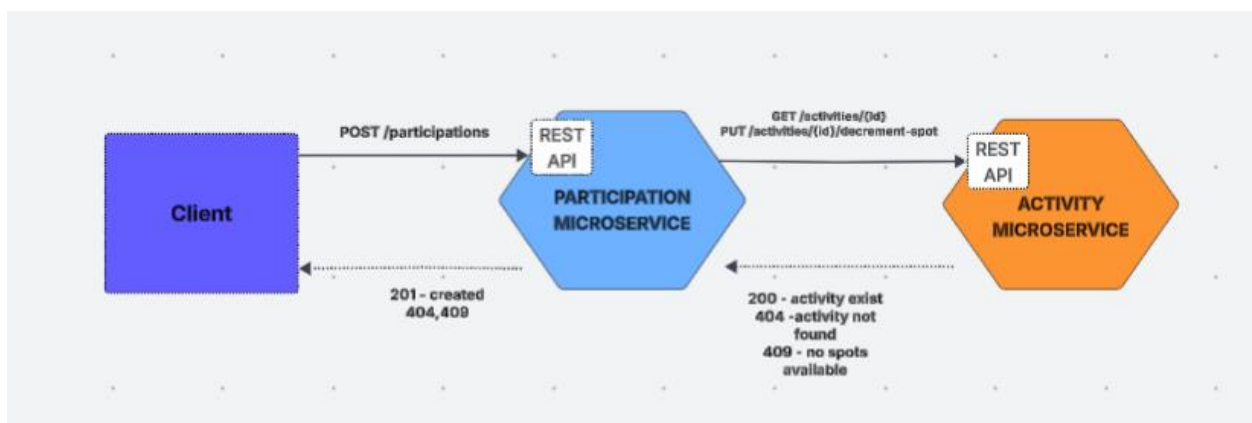
Zadatak 5

U nastavku se nalaze dijagrami sinhronne komunikacije.

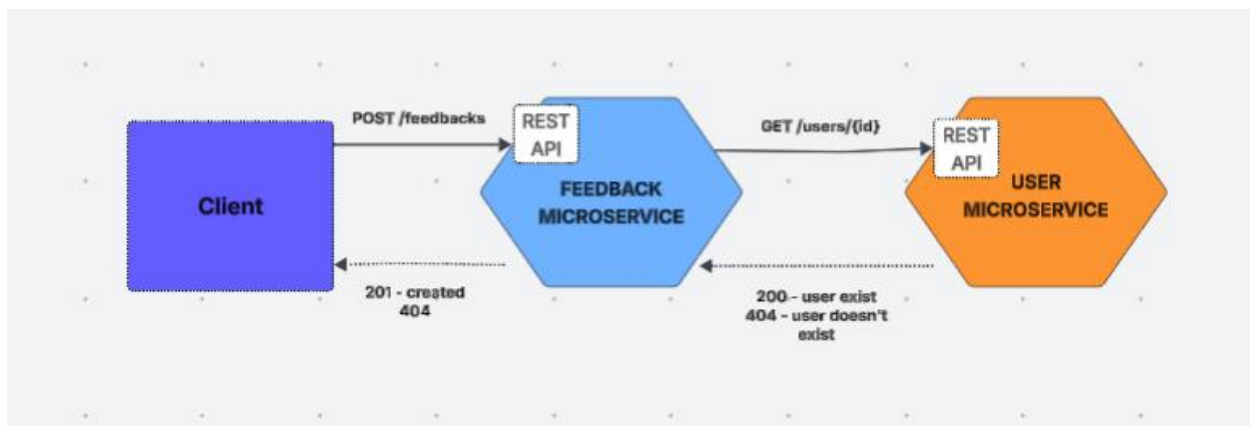
1. Komunikacija između **participation-service** i **user-service** prilikom validacije korisnika prije kreiranja prijave za aktivnost:



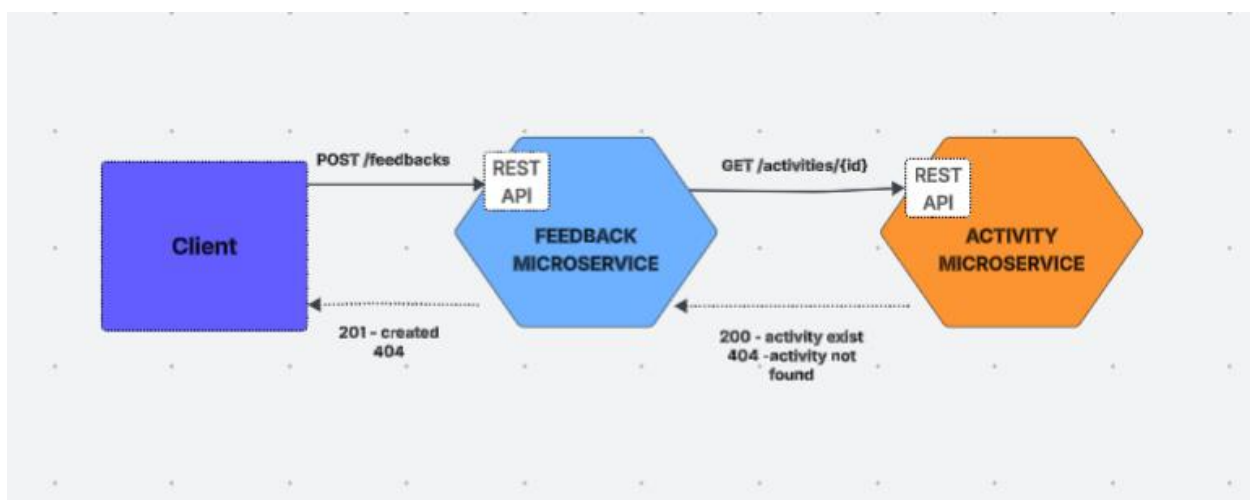
2. Komunikacija između **participation-service** i **activity-service** prilikom provjere da li aktivnost postoji i ima mjesta:



3. Komunikacija između **feedback-service** i **user-service** prilikom validacije korisnika koji ostavlja feedback:



4. Komunikacija između **feedback-service** i **activity-service** prilikom provjere da li aktivnost postoji:



Zadatak 7

Sigurnosno rješenje za sistem baziran na mikroservisnoj arhitekturi

1. Uvod i pregled istraživanja

Sigurnost u mikroservisnoj arhitekturi je inherentno složenija u poređenju sa tradicionalnim monolitnim aplikacijama. Distribuirana priroda sistema, gdje svaki mikroservis predstavlja potencijalnu ulaznu tačku i nezavisnu komponentu, zahtijeva

pažljivo planiranje i implementaciju sigurnosnih mehanizama. Osnovni cilj je osigurati da samo ovlašteni korisnici i servisi mogu pristupiti resursima, istovremeno održavajući skalabilnost, fleksibilnost i performanse.

Istraživanje različitih rješenja za sigurnost mikroservisa otkrilo je nekoliko ključnih obrazaca i tehnologija:

- **API Gateway kao Kontrolna Tačka:** Većina modernih mikroservisnih arhitektura koristi API Gateway kao centralnu tačku ulaska za sve eksterne zahtjeve. Ova komponenta je idealno mjesto za implementaciju prve linije odbrane, uključujući autentifikaciju, validaciju tokena i grubo-zrnastu autorizaciju. Time se rasterećuju pojedinačni mikroservisi od odgovornosti za ove zadatke.
- **Token-Based Autentifikacija:** Umjesto tradicionalnih sesija (koje su "stateful" i otežavaju horizontalnu skalabilnost), distribuirani sistemi preferiraju tokene. Ovi tokeni sadrže sve potrebne informacije za autentifikaciju i autorizaciju unutar samog tokena ("stateless" pristup). Najpopularniji standard za ovo je **JSON Web Token (JWT)**, zbog svoje kompaktnosti, samostalnosti i mogućnosti digitalnog potpisivanja.
- **Decentralizovana Autorizacija:** Iako se autentifikacija često centralizuje na API Gatewayu, autorizacija (provjera specifičnih dozvola za pristup resursima) se uglavnom prepušta pojedinačnim mikroservisima. Ovo omogućava svakom servisu da implementira granularnu kontrolu pristupa specifičnu za njegove resurse, bez stvaranja uskog grla na centralnom serveru.
- **Komunikacija među servisima (Service-to-Service Security):** Izuzetno je važno osigurati i komunikaciju između samih mikroservisa. Ne smije se pretpostaviti da je interna mreža uvijek sigurna. Stoga se često koriste slični token-based mehanizmi ili certifikati za međuservisnu autentifikaciju i autorizaciju.
- **Rukovanje stanjem tokena:** Budući da su JWT-ovi po prirodi "stateless", pitanja poput "logouta" i "invalidacije" zahtijevaju posebna rješenja, obično kroz kombinaciju kratkog trajanja tokena, "refresh" tokena i crnih lista (blacklists).

Na osnovu ovih uvida, predloženo rješenje fokusira se na robustan i skalabilan pristup sigurnosti, uzimajući u obzir specifičnosti aplikacije za upravljanje volonterima koja uključuje servise kao što su **user-service**, **feedback-service**, **participation-service**, **system-events-service**, te infrastrukturne komponente poput **api-gateway**, **eureka-server** i **config-server**.

2. Predloženo Sigurnosno Rješenje

Predloženo rješenje za sigurnost aplikacije bazirane na mikroservisnoj arhitekturi je **autentifikacija i autorizacija bazirana na JWT (JSON Web Token) standardu, s API Gatewayom kao centralnom tačkom za autentifikaciju eksternih zahtjeva.**

Obrazloženje Rješenja:

Ovo rješenje kombinuje najbolje prakse centralizovanog i decentralizovanog pristupa sigurnosti:

- **Centralizovana Autentifikacija:** API Gateway preuzima ulogu "policajca na ulazu", obrađujući sve početne autentifikacijske provjere. Time se postiže jedinstvena tačka ulaska i rasterećuju pozadinski servisi.
- **Decentralizovana Autorizacija putem JWT-a:** Nakon početne validacije, JWT se proslijeđuje mikroservisima. Svaki mikroservis može samostalno validirati integritet i valjanost tokena, a zatim koristiti informacije (claims) unutar tokena (npr. uloge korisnika) za donošenje odluka o autorizaciji za svoje resurse. Ovo smanjuje latenciju i povećava skalabilnost sistema, jer servisi ne moraju komunicirati s centralnim autorizacijskim serverom za svaku provjeru.
- **Stateless priroda JWT-a:** Eliminacija serverski pohranjenih sesija pojednostavljuje horizontalno skaliranje mikroservisa, jer se ne mora brinuti o replikaciji sesija ili ljepljivim sesijama.
- **Standardizacija:** Korištenje JWT-a osigurava interoperabilnost i olakšava integraciju s različitim klijentskim aplikacijama (web, mobilne, desktop).

3. Detaljni Odgovori na Pitanja

a. Da li će API Gateway voditi računa o autentifikaciji?

DA. API Gateway će biti primarna tačka za autentifikaciju svih eksternih zahtjeva.

- **Proces autentifikacije:**
 1. Klijent (npr. web aplikacija, mobilna aplikacija) šalje kredencijale (korisničko ime i lozinku) na `/auth/login` endpoint koji je rutiran kroz API Gateway do **User Service**-a (koji će biti odgovoran za autentifikaciju).
 2. **User Service** provjerava kredencijale u bazi podataka i, ukoliko su validni, generiše **JWT Access Token** i **Refresh Token**.
 3. **User Service** vraća tokene API Gatewayu, koji ih proslijeđuje klijentu.
 4. Svi budući zahtjevi klijenta prema bilo kojem mikroservisu (preko API Gatewaya) moraju sadržavati JWT Access Token u **Authorization** headeru (npr. **Bearer <JWT_TOKEN>**).

5. API Gateway će imati konfigurisan filter (npr. Spring Cloud Gateway Filter) koji presreće sve dolazne zahtjeve, validira JWT (provjerava potpis, istek tokena i eventualno prisustvo na crnoj listi).
6. Ukoliko je token validan, API Gateway će proslijediti zahtjev odgovarajućem mikroservisu, često dodajući korisničke informacije (kao što su ID korisnika i uloge) u zaglavlja zahtjeva (npr. `X-User-ID`, `X-User-Roles`).
7. Ako token nije validan (npr. istekao je, potpis je neispravan, ili je na crnoj listi), zahtjev se odbija direktno na API Gatewayu sa odgovarajućim HTTP status kodom (npr. 401 Unauthorized ili 403 Forbidden).

Prednosti ovakvog pristupa:

- **Centralizovana kontrola pristupa:** Jedna tačka ulaska za sigurnosne provjere.
- **Rasterećenje mikroservisa:** Pojedinačni servisi ne moraju implementirati punu logiku autentifikacije.
- **Smanjena površina napada:** Samo API Gateway je izložen javno, dok su ostali mikroservisi u privatnoj mreži.

b. Da li ćete koristiti tokene? (JWT ili neki drugi)

DA, koristit ćemo JSON Web Token (JWT).

JWT je odabran zbog svoje "stateless" prirode, što je idealno za distribuirane sisteme i skalabilnost.

- **Struktura JWT-a:** JWT se sastoji od tri dijela, odvojena tačkama (.):
 1. **Header:** Sadrži tip tokena (JWT) i algoritam hashiranja koji se koristi za potpisivanje (npr. HS256, RS256).
 2. **Payload (Claims):** Sadrži "claims" (tvrdnje) o entitetu (obično korisniku) i dodatne metapodatke. To su key-value parovi koji mogu uključivati:
 - Standardne (registrovane) claimove: `iss` (issuer), `exp` (expiration time), `sub` (subject), `aud` (audience), `jti` (JWT ID - jedinstveni identifikator tokena).
 - Javne claimove: Predefinisani, ali opcioni.
 - Privatne claimove: Custom claims, kao što su `userId`, `roles`, `email`, koji su specifični za aplikaciju.
 3. **Signature:** Kreira se digitalnim potpisivanjem Header-a, Payload-a i tajnog ključa (ili privatnog ključa u slučaju asimetričnog šifrovanja). Potpis osigurava integritet tokena (da sadržaj nije mijenjan) i autentičnost (da je token izdat od strane pouzdanog izvora).

Prednosti JWT-a:

- **Stateless (bez stanja):** Nema potrebe za pohranjivanjem sesija na serveru, što značajno pojednostavljuje skaliranje i distribuciju.
- **Samostalnost:** Sve potrebne informacije za autorizaciju (osim osjetljivih podataka) su sadržane unutar samog tokena. Mikroservisi mogu validirati token bez direktne komunikacije s bazom podataka ili autorizacijskim serverom za svaki zahtjev.
- **Standardizovan:** JWT je otvoren standard (RFC 7519), što osigurava interoperabilnost među različitim platformama i tehnologijama.
- **Digitalno potpisan:** Potpis garantuje integritet tokena, čineći ga otpornim na neovlaštene izmjene.

c. Kako ćete realizovati role za korisnike? Permisije za role?

Realizacija rola:

- **Pohrana Rola:** Role (npr. **ADMIN**, **VOLUNTEER**, **ORGANIZER**, **GUEST**) će biti pohranjene u bazi podataka unutar **User Service**-a. Svakom korisniku će biti dodijeljena jedna ili više rola.
- **Uključivanje u JWT:** Kada se JWT generiše tokom procesa prijave, imena rola dodijeljenih korisniku će biti uključena u **Payload** JWT-a kao "custom claim" (npr. `{ "userId": "...", "roles": ["VOLUNTEER", "ORGANIZER"], ... }`).

Permisije za role (Autorizacija): Za finu granularnu kontrolu pristupa, koristiti će se kombinacija uloga i permisija implementiranih na nivou mikroservisa.

- **Uloge za grubu autorizaciju:** Role iz JWT-a će se koristiti za grubu autorizaciju (npr. "samo ADMIN može pristupiti **/admin** endpointima"). Ovo se može implementirati na API Gatewayu i/ili na pojedinačnim mikroservisima.
- **Permisije za finu granularnu autorizaciju:**
 - **Policy-Based Access Control (PBAC):** Preporučuje se implementacija PBAC-a unutar svakog mikroservisa. Svaki mikroservis će definisati politike autorizacije specifične za njegove resurse. Ove politike će uzeti u obzir:
 - **Uloge korisnika:** Izvučene iz JWT-a.
 - **ID korisnika:** Također izvučen iz JWT-a (npr. **userId**).
 - **Kontekst zahtjeva:** Npr. ID resursa koji se pristupa (da li korisnik smije uređivati svoj događaj, a ne tuđi).

- **Primjer implementacije (Spring Security):** U Spring Boot mikroservisima, ovo se može postići pomoću `@PreAuthorize` anotacija na nivoima kontrolera ili servisa, koristeći SpEL (Spring Expression Language) za kompleksne provjere:
 - `@PreAuthorize("hasRole('ADMIN')")`: Samo korisnici s ulogom ADMIN.
 - `@PreAuthorize("hasAnyRole('VOLUNTEER', 'ORGANIZER')")`: Korisnici s ulogom VOLUNTEER ili ORGANIZER.
 - `@PreAuthorize("hasRole('ORGANIZER') and @eventService.isOwner(#eventId, principal.userId)")`: Primjer gdje ORGANIZER može pristupiti samo ako je vlasnik događaja (koristeći servis za provjeru vlasništva).

d. Centralizovana ili decentralizovana autorizacija? - obrazložiti prednosti i mane oba pristupa u slučaju vašeg projekta

Za našu aplikaciju predlažem **HIBRIDNI PRISTUP** sa tendencijom ka **DECENTRALIZOVANOJ** autorizaciji.

- **Centralizovana Autentifikacija:** Kao što je prethodno navedeno, autentifikacija se centralizuje na API Gatewayu. Ovo je centralizovani dio našeg rješenja.
- **Decentralizovana Autorizacija:** Nakon što API Gateway validira JWT i proslijedi zahtjev, svaki pojedinačni mikroservis je odgovoran za autorizaciju specifičnih operacija na svojim resursima.

Prednosti hibridnog pristupa (sa decentralizovanom autorizacijom):

- **Skalabilnost:** Svaki mikroservis donosi odluke o autorizaciji nezavisno, bez potrebe za komunikacijom s centralnim autorizacijskim serverom za svaki zahtjev. Ovo smanjuje opterećenje na autorizacijski server i značajno smanjuje latenciju, što je ključno za aplikacije s visokim prometom.
- **Fleksibilnost i Autonomija Servisa:** Svaki mikroservis može definisati i implementirati vlastita, specifična pravila autorizacije koja su najrelevantnija za njegove domene i resurse. Ovo podržava autonomiju mikroservisa.
- **Otpornost (Resilience):** Ako centralni autorizacijski server (tj. `User Service` koji izdaje tokene) postane nedostupan nakon što su tokeni već izdati, operativni mikroservisi i dalje mogu funkcionisati i procesirati zahtjeve (dok su tokeni validni), jer ne ovise o stalnoj vezi s njim za svaku autorizaciju.

Mane (Decentralizovana autorizacija):

- **Dupliranje logike validacije tokena:** Svaki mikroservis mora imati kod za validaciju JWT-a (provjera potpisa, isteka). Ovo se ublažava korištenjem zajedničke biblioteke/modula (npr. `common-dto` ili posebnog `security-lib`) koja sadrži tu logiku.
- **Izazovi s invalidacijom tokena:** Trenutna invalidacija tokena (npr. prilikom logouta ili kompromitacije) je složenija jer svaki servis ne provjerava centralno stanje tokena za svaki zahtjev. Ovo se rješava kombinacijom kratkog trajanja tokena, refresh tokena i centralne crne liste (što je objašnjeno dalje).

Centralizovana autorizacija (potpuna):

- **Prednosti:** Sva logika autorizacije na jednom mjestu, što olakšava globalno upravljanje pravilima i reviziju.
- **Mane:** Stvara usko grlo (`bottleneck`) jer svaki autorizacijski zahtjev mora proći kroz centralnu tačku. Povećava latenciju sistema. Predstavlja jednu tačku kvara (`single point of failure`). Nije skalabilno i nije u skladu s filozofijom mikroservisa.

Za naš projekat, hibridni pristup je optimalan jer iskorištava prednosti oba modela, osiguravajući efikasnost i sigurnost.

e. Autorizacija između mikroservisa? Da li je potrebna? Da li je moguće uraditi bez autorizacije na svakom mikroservisu? Pristup iz vana pojedinačnim mikroservisima?

DA, autorizacija između mikroservisa je potrebna i visoko preporučljiva.

- **Zašto je potrebna:**
 - **Princip najmanje privilegije:** Svaki servis bi trebao imati samo one dozvole koje su mu neophodne za obavljanje njegove funkcije.
 - **Sprečavanje lateralnog kretanja napadača:** Ako napadač kompromituje jedan mikroservis, autorizacija između servisa sprječava ga da se slobodno kreće i pristupa drugim servisima bez provjere.
 - **Diferenciran pristup:** Različiti servisi mogu imati različite nivoe pristupa podacima ili funkcionalnostima drugih servisa (npr. `Feedback Service` možda smije samo čitati podatke iz `User Service`-a, ali ne smije ih mijenjati).
- **Kako realizovati autorizaciju između mikroservisa:**
 - **Prosljeđivanje korisničkog JWT-a:** API Gateway, nakon validacije korisničkog JWT-a, prosljeđuje ga dalje pozvanom mikroservisu. Pozvani mikroservis ponovo validira integritet tokena i koristi informacije o

korisniku (ID, uloge) za donošenje odluka o autorizaciji. Ovo je primarni mehanizam za "on behalf of" korisničke zahtjeve.

- **Service-to-Service Token / Service Accounts:** Za scenarije gdje ne postoji direktan korisnički zahtjev (npr. interni asinhroni procesi, background poslovi), mikroservisi mogu imati svoje "service accounts" i koristiti "service-to-service" tokene. **User Service** (ili poseban **Auth Service**) može izdavati kratkotrajne tokene za specifične servise. Mikroservis **A** bi, prije pozivanja mikroservisa **B**, dobio takav token i poslao ga. Mikroservis **B** bi validirao taj token i provjerio da li servis **A** ima dozvolu za tu operaciju.
- **MTLS (Mutual TLS):** Za najviši nivo sigurnosti, može se implementirati Mutual TLS, gdje se oba klijenta i server autentifikuju koristeći X.509 certifikate. Ovo osigurava autentifikaciju na transportnom sloju.
- **Da li je moguće uraditi bez autorizacije na svakom mikroservisu?** Teoretski da, ali **praktično i sigurnosno NEPREPORUČLJIVO**. Ako je API Gateway jedina tačka autentifikacije i mikroservisi slijepo vjeruju svim dolaznim zahtjevima (bez ponovne validacije tokena ili autorizacije), to stvara ogroman sigurnosni rizik. Svaka kompromitacija gatewaya ili zaobilaženje gatewaya omogućilo bi neograničen pristup internim servisima.
- **Pristup iz vana pojedinačnim mikroservisima? APSULUTNO NE.** Pojedinačni mikroservisi **NIKADA ne smiju biti direktno izloženi javno**. Svi eksterni zahtjevi moraju ići isključivo preko API Gatewaya. Mikroservisi trebaju biti smješteni u privatnoj mreži ili zaštićeni robustnim firewall pravilima koja dozvoljavaju pristup samo API Gatewayu i eventualno drugim internim servisima. Ovo je fundamentalni princip sigurnosti u mikroservisnoj arhitekturi.

f. Logout i tokeni? Da li će token ostati validan i nakon logouta?

Da, JWT Access Token će po defaultu ostati validan i nakon logouta.

- **Stateless priroda:** Budući da su JWT-ovi "stateless", server (ili mikroservisi) ne održavaju nikakvo stanje o izdatim tokenima. Kada korisnik izvrši operaciju "logout", klijentska aplikacija jednostavno odbacuje token. Server ne dobija nikakvu obavijest o tome i token ostaje tehnički validan sve dok ne istekne njegovo predviđeno trajanje.
- Ovo je izazov kod JWT-a koji se mora riješiti kroz dodatne mehanizme, kao što je invalidacija.

g. Invalidacija tokena? Da li je potrebna? - obrazložiti

DA, invalidacija tokena je nužna i preporučljiva, pogotovo u scenarijima koji zahtijevaju trenutni opoziv pristupa.

Zašto je potrebna invalidacija:

- **Eksplícitan Logout:** Iako je JWT stateless, korisničko iskustvo i sigurnost zahtijevaju da se pristup odmah opozove nakon što se korisnik odjavi.
- **Kompromitacija tokena:** Ako se JWT Access Token ukrade (npr. putem XSS napada ili socijalnog inženjeringa), potrebna je mogućnost trenutne invalidacije kako bi se minimizovala potencijalna šteta.
- **Promjena korisničkih privilegija:** Ako se korisniku promijene uloge ili dozvole, ili ako se njegov račun deaktivira, želite da bilo koji aktivni tokeni sa starim privilegijama odmah prestanu biti validni.
- **Resetovanje lozinke:** Nakon resetovanja lozinke, trebalo bi invalidirati sve aktivne tokene za tog korisnika kako bi se osiguralo da stari, potencijalno kompromitovani tokeni, ne mogu biti korišteni.

Kako realizovati invalidaciju tokena:

1. **Kratko trajanje Access Tokena:** Ovo je prva linija odbrane. Postavljanjem Access Tokena na vrlo kratko trajanje (npr. 5-15 minuta) dramatično se smanjuje prozor ranjivosti.
2. **Korištenje Refresh Tokana:** Umjesto da se Access Tokeni čine dugotrajnim, koristi se dugotrajni Refresh Token. Refresh Token su "stateful" (pohranjuju se u bazi podataka na Auth Service-u) i mogu se opozvati (izbrisati iz baze podataka) prilikom logouta, resetovanja lozinke, ili ako se otkrije sumnjiva aktivnost.
3. **Blacklist (Lista opoziva):** Za trenutnu invalidaciju Access Tokena (npr. prilikom logouta ili sumnjivih aktivnosti), koristi se centralizovana crna lista.
 - Kada token treba biti invalidiran, njegov jedinstveni ID (**jti** claim iz JWT-a) se dodaje na crnu listu (npr. u Redis, zbog svoje brzine).
 - Crna lista pohranjuje **jti** i originalno vrijeme isteka tokena.
 - Prilikom svake validacije tokena na API Gatewayu (ili mikroservisima), pored provjere potpisa i isteka, provjerava se i da li je **jti** tokena prisutan na crnoj listi. Ako jeste, token se odbija, bez obzira na njegovo originalno vrijeme isteka.
 - Unosi na crnoj listi se automatski brišu kada istekne originalno trajanje tokena.

h. Trajanje tokena i Refresh Token

- **Access Token (JWT):**
 - **Trajanje: Kratkotrajno.** Preporučuje se trajanje od **5 do 15 minuta**.

- **Obrazloženje:** Ovako kratko trajanje minimizira rizik od zloupotrebe kompromitovanog tokena. Čak i ako napadač ukrade token, ima vrlo kratak vremenski okvir prije nego što token istekne.
- **Refresh Token:**
 - **Trajanje: Dugotrajno.** Trajanje može biti **od nekoliko sati do nekoliko dana ili sedmica**, u zavisnosti od nivoa sigurnosti i korisničkog iskustva koje želite postići.
 - **Obrazloženje:** Refresh Token omogućava korisniku da ostane prijavljen u aplikaciji bez potrebe za ponovnim unosom korisničkog imena i lozinke. Kada Access Token istekne, klijent koristi Refresh Token da zatraži novi Access Token od **User Service**-a.
 - **Sigurnost Refresh Tokena:** Refresh Tokeni su "stateful" i pohranjuju se u bazi podataka **User Service**-a. Moraju biti zaštićeni i jedinstveni za svaku sesiju/uređaj. Mogu se opozvati (izbrisati) u bilo kojem trenutku (npr. prilikom logouta, sumnjive aktivnosti ili resetovanja lozinke). Klijent bi trebao pohranjivati Refresh Token na najsigurniji mogući način (npr. HTTP-Only cookies za web, Keychain/Keystore za mobilne).

Proces korištenja Access i Refresh Tokena:

1. Korisnik se prijavi putem **User Service**-a, dobija **Access Token** (kratkotrajan) i **Refresh Token** (dugotrajan).
2. Klijent koristi Access Token za sve zahtjeve prema API-ju.
3. Kada Access Token istekne (ili je blizu isteka), klijent šalje Refresh Token **User Service**-u putem posebnog endpointa (npr. **/auth/refresh**).
4. **User Service** validira Refresh Token (provjerava da li je validan, nije opozvan, itd.). Ako je validan, izdaje novi Access Token i opcionalno novi Refresh Token (za rotaciju refresh tokena, što dodatno povećava sigurnost).
5. Ako se korisnik odjavi, Refresh Token se opoziva u **User Service**-u (briše iz baze podataka).

i. Pristup API-u sa mobilnih uređaja? Da li je potrebno? - obrazložiti

DA, pristup API-u sa mobilnih uređaja je vrlo vjerovatno potreban i predstavlja ključnu funkcionalnost za modernu aplikaciju.

Obrazloženje:

- **Mobilna dominacija:** Danas veliki broj korisnika pristupa internet uslugama putem mobilnih uređaja. Pružanje funkcionalne mobilne aplikacije ili pristupa putem

mobilnog web pretraživača značajno povećava doseg vaše aplikacije i poboljšava korisničko iskustvo.

- **Jedinstveni API:** Korištenjem RESTful API-ja, isti set endpointa koji služi web aplikaciji može lako biti iskorišten i od strane nativnih mobilnih aplikacija (Android/iOS) ili mobilnih web aplikacija. To smanjuje dupliranje posla i pojednostavljuje održavanje.
- **JWT i mobilni uređaji:** JWT je izuzetno pogodan za mobilne aplikacije jer je "stateless". Token se može sigurno pohraniti na uređaju (npr. u **Keychain** na iOS-u, **Keystore** na Androidu ili **SharedPreferences** uz dodatnu enkripciju), bez potrebe za serverskim sesijama koje bi bile kompleksne za upravljanje na mobilnim uređajima.
- **Sigurnosni aspekti na mobilnim uređajima:**
 - **Sigurna pohrana tokena:** Ključno je osigurati da se tokeni sigurno pohranjuju na mobilnom uređaju.
 - **Uvijek HTTPS:** Sva komunikacija između mobilne aplikacije i API-ja mora ići preko HTTPS-a kako bi se zaštitila razmjena podataka.
 - **OAuth 2.0 sa PKCE:** Za robusniju i sigurniju autentifikaciju na mobilnim uređajima, preporučuje se korištenje OAuth 2.0 sa Proof Key for Code Exchange (PKCE) ekstenzijom. PKCE pruža dodatni sloj sigurnosti protiv "authorization code interception" napada.

Zadatak 8

Sigurnosno rješenja kroz Postman

1. Autentifikacija (Prijava)

1.1. Registracija novog korisnika

- Svrha: Kreirati korisnike sa različitim ulogama za testiranje autorizacije

The screenshot displays the Postman interface for a REST client. At the top, a POST request is configured to the URL `http://localhost:8082/api/auth/signup`. The 'Body' tab is selected, showing a JSON payload for user registration. The response status is `200 OK` with a response time of 5.81 s and a body size of 468 B. The response body contains a success message.

Request:

```
POST http://localhost:8082/api/auth/signup
```

Body (JSON):

```
{  "username": "volunteer1",  "email": "volunteer1@example.com",  "password": "Password123!",  "firstName": "Volonter",  "lastName": "Jedan",  "profilePicture": "http://example.com/vol1.jpg",  "roles": ["volunteer"]}
```

Response:

```
{  "message": "Korisnik uspješno registrovan!"}
```


POST

http://localhost:8082/api/auth/signup

Send

Params Auth Headers (10) **Body** Scripts Settings

Cookies

raw

JSON

Beautify

```
1 {  
2   "username": "organizer1",  
3   "email": "organizer1@example.com",  
4   "password": "Password123!",  
5   "firstName": "Organizator",  
6   "lastName": "Jedan",  
7   "profilePicture": "http://example.com/org1.jpg",  
8   "roles": ["organizer"]  
9 }
```

Body

200 OK • 206 ms • 468 B •

{ } JSON

Preview

Visualize

```
1 {  
2   "message": "Korisnik uspješno registrovan!"  
3 }
```

POST

http://localhost:8082/api/auth/signup

Send

Params

Auth

Headers (10)

Body ●

Scripts

Settings

Cookies

raw

JSON

Beautify

```
1  {
2    "username": "admin1",
3    "email": "admin1@example.com",
4    "password": "Password123!",
5    "firstName": "Administrator",
6    "lastName": "Jedan",
7    "profilePicture": "http://example.com/adm1.jpg",
8    "roles": ["admin"]
9  }
```

Body

200 OK

344 ms

468 B

{}

JSON

Preview

Visualize

```
1  {
2    "message": "Korisnik uspješno registrovan!"
3  }
```

```
"message": "Greška: could not execute statement [ERROR: duplicate key value
violates unique constraint \"uk_r43af9ap4edm43mmqt01oddj6\"\\n  Detail: Key
(username)=(admin1) already exists.] [insert into users
(created_at,email,first_name,last_name,password_hash,profile_picture,updated_at,userna
me) values (?,?,?,?,?,?,?,?)] ; SQL [insert into users
(created_at,email,first_name,last_name,password_hash,profile_picture,updated_at,userna
me) values (?,?,?,?,?,?,?,?)] ; constraint [uk_r43af9ap4edm43mmqt01oddj6] "
```

1.2 Prijava korisnika i dobijanje JWT tokena

POST

http://localhost:8082/api/auth/signin

Send

Params Auth Headers (10) **Body** Scripts Settings

Cookies

raw

JSON

Beautify

```
1  {
2    "username": "volunteer1",
3    "password": "Password123!"
4  }
```

Body

200 OK

944 ms

663 B

{ } JSON

Preview

Visualize

```
1  {
2    "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ2b2x1bnRlZXIiOiwiawF0IjoxNzQ5MDQxMDI3LCJleHAiOiJl3NDk5Mjc0Mjd9.NaqDVS-ddBx-REnrksTj3bY6HM6KsEdmPpLYESnQTeQ",
3    "type": "Bearer",
4    "id": 5,
5    "username": "volunteer1",
6    "email": "volunteer1@example.com",
7    "roles": []
8  }
```

POST

▼

http://localhost:8082/api/auth/signin

Send

▼

ParamsAuthHeaders (10)Body●ScriptsSettingsCookies

raw▼JSON▼Beautify

```
1  {
2    "username": "organizer1",
3    "password": "Password123!"
4  }
```



Body▼🕒200 OK • 251 ms • 663 B • 🌐⋮

{{}JSON▼▶Preview🔗Visualize▼

⌵⌵🔍📄🔗

```
1  {
2    "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJvcmdhbml6ZXIiOiwiWF0IjoxNzQ5MDQxMDkxLCJleHAiOiE3NDk5Mjc0OTF9.ojBC6sXYVZslaFbvuy4pR9FA12fQA3YZL60ue4fv05k",
3    "type": "Bearer",
4    "id": 6,
5    "username": "organizer1",
6    "email": "organizer1@example.com",
7    "roles": []
8  }
```

POST http://localhost:8082/api/auth/signin

Send

Params Auth Headers (10) **Body** Scripts Settings Cookies

raw JSON Beautify

```
1 {
2   "username": "admin1",
3   "password": "Password123!"
4 }
```

Body 200 OK • 197 ms • 650 B •

{ } JSON Preview Visualize

```
1 {
2   "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJhZG1pbjEiLCJpYXQiOiJlE3NDk4NDExMjIsImV4cCI6MTc0OTkyNzUyMn0uYnZUyMn0.C-8m9G0EA2wnwtJAAtNJvQnQ_HnZvoqgM75JN9tPuVqs",
3   "type": "Bearer",
4   "id": 7,
5   "username": "admin1",
6   "email": "admin1@example.com",
7   "roles": []
8 }
```

1.2.1 Testiranje pogrešne autentifikacije

POST

http://localhost:8082/api/auth/signin

Send

Params Auth Headers (10) **Body** Scripts Settings

Cookies

raw

JSON

Beautify

```
1 {  
2   "username": "nepostojecikorisnik",  
3   "password": "Password123!"  
4 }
```

Body

401 Unauthorized

227 ms

488 B



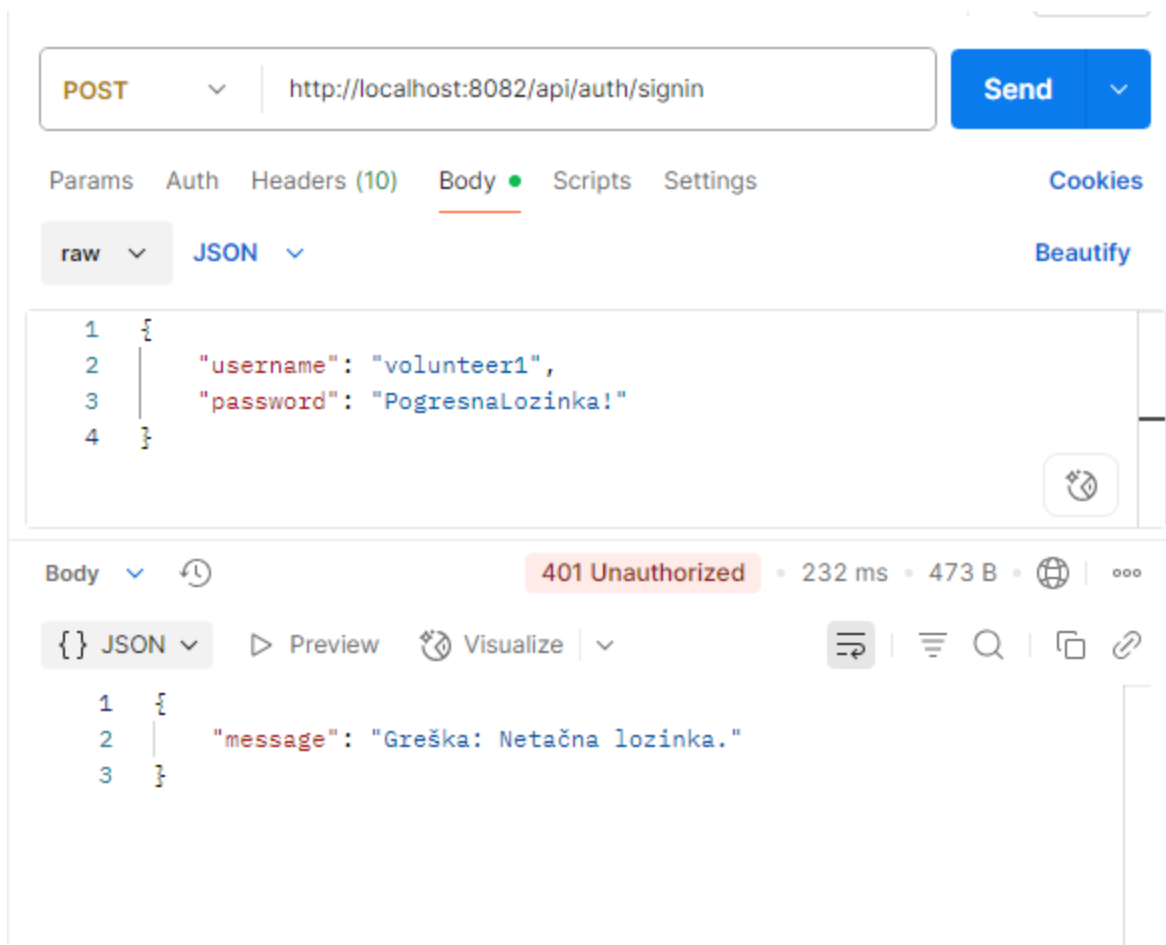
{ } JSON

Preview

Visualize

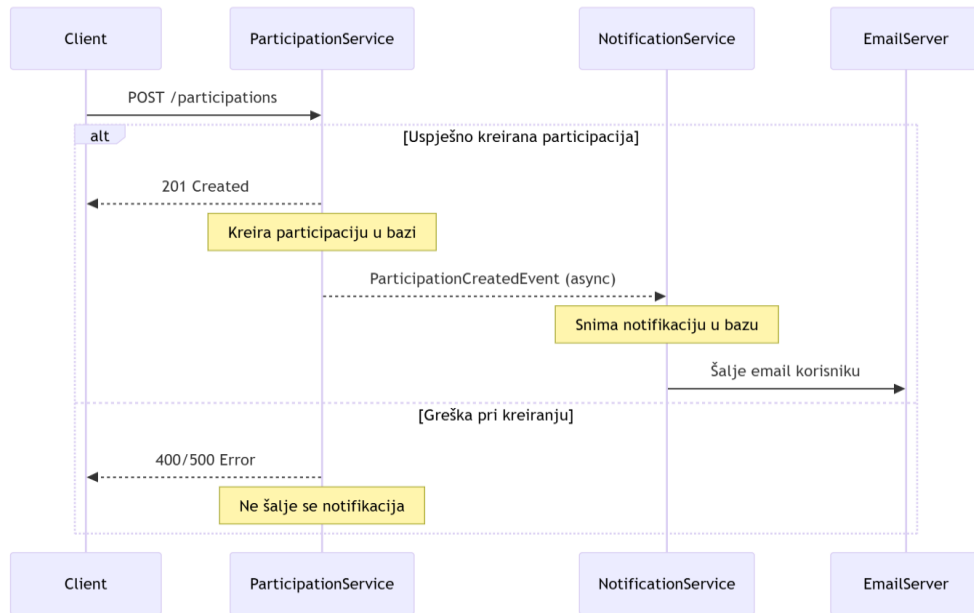


```
1 {  
2   "message": "Greška: Korisničko ime nije pronađeno."  
3 }
```

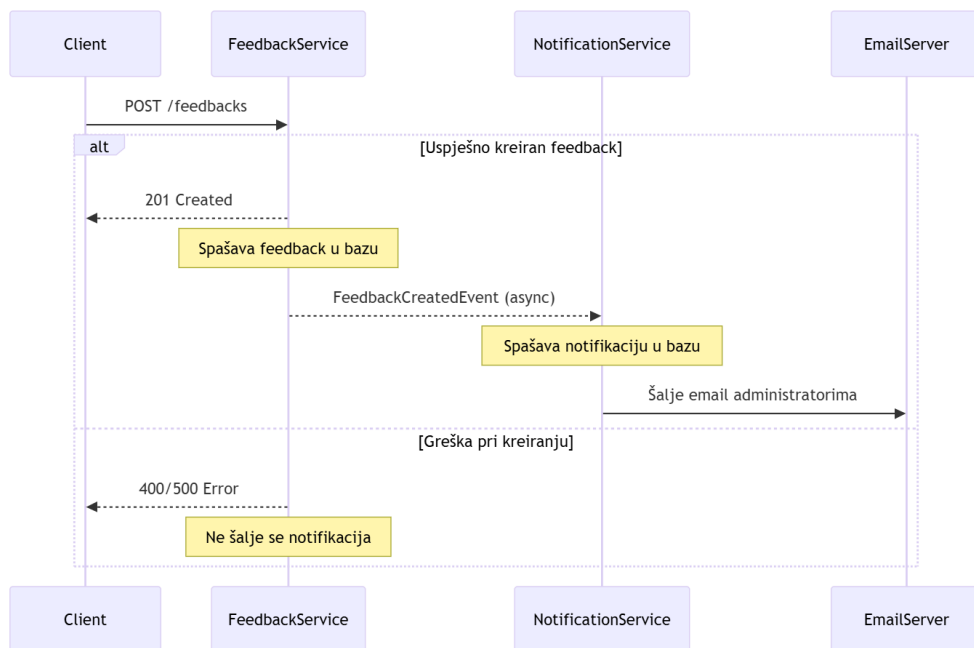


U nastavku se nalaze dijagrami asinhronne komunikacije:

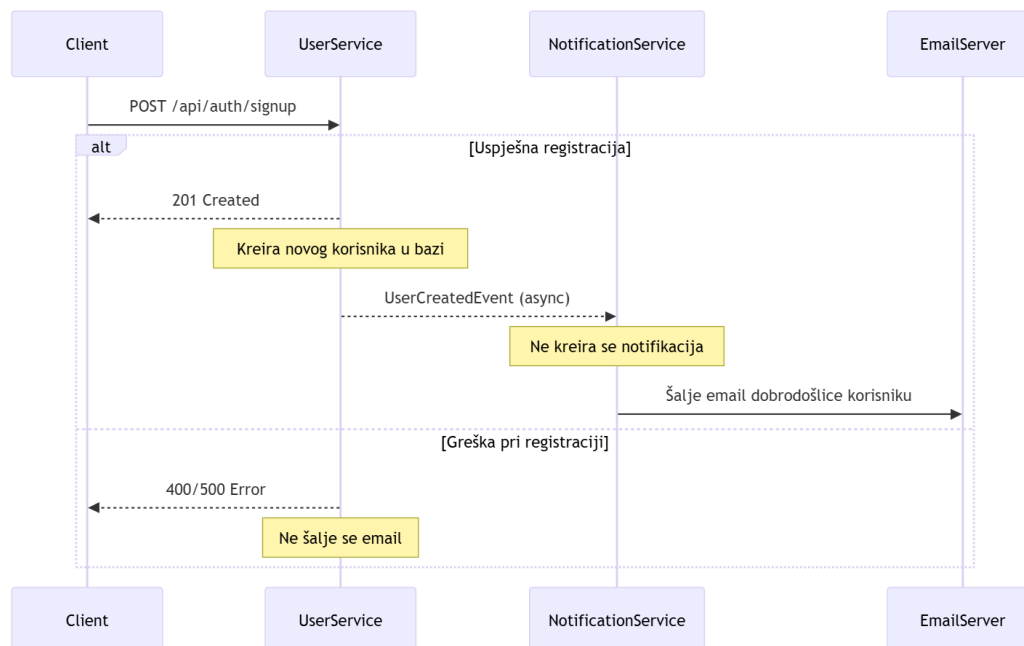
1. Komunikacija između participation servisa i notification servisa prilikom kreiranja prijave za aktivnost:



2. Komunikacija između feedback servisa i notification servisa prilikom kreiranja feedbacka za aktivnost:



3. Komunikacija između user servisa i notification servisa prilikom registracije:



Zadaci 2 i 3 u zadatku 10 nisu implementirani zbog vremenskih ograničenja i prioritizacije funkcionalnosti sistema.

Tokom razvoja, fokus je bio na implementaciji i testiranju ključnih poslovnih funkcionalnosti mikroservisa (kao što su registracija korisnika, prijava na aktivnosti, slanje notifikacija i feedback sistem).

Postavljanje Docker okruženja i **docker-compose** fajla zahtijeva dodatno vrijeme za konfiguraciju, optimizaciju memorije i međusobno povezivanje servisa.